LECTURE · HMM APPLICATIONS

Hidden Markov Model Applications

From K-Means Clustering to Market Regimes and Informed Trading

K-Means → Markov chain → forward, Viterbi, Baum-Welch → stock prediction → PIN / VPIN

Figures in this deck are based on the original lecture slides.
Worked examples: weather and daily activity; KOSPI returns; EUR/USD futures order flow.

Clustering, sequences and regimes share one EM idea

Three ideas carry this lecture

1

K-Means groups points without time order

Candles are clustered by shape; the order in which they occur is ignored.

2

An HMM adds time through a Markov chain

Hidden states move with transition probabilities and emit what we observe.

3

The same algorithms run on market data

Forward, Viterbi and Baum-Welch turn KOSPI returns and order flow into regime signals.

Both K-Means and Baum-Welch alternate an E-step and an M-step.

01

PART 1 · CLUSTERING

K-Means Clustering

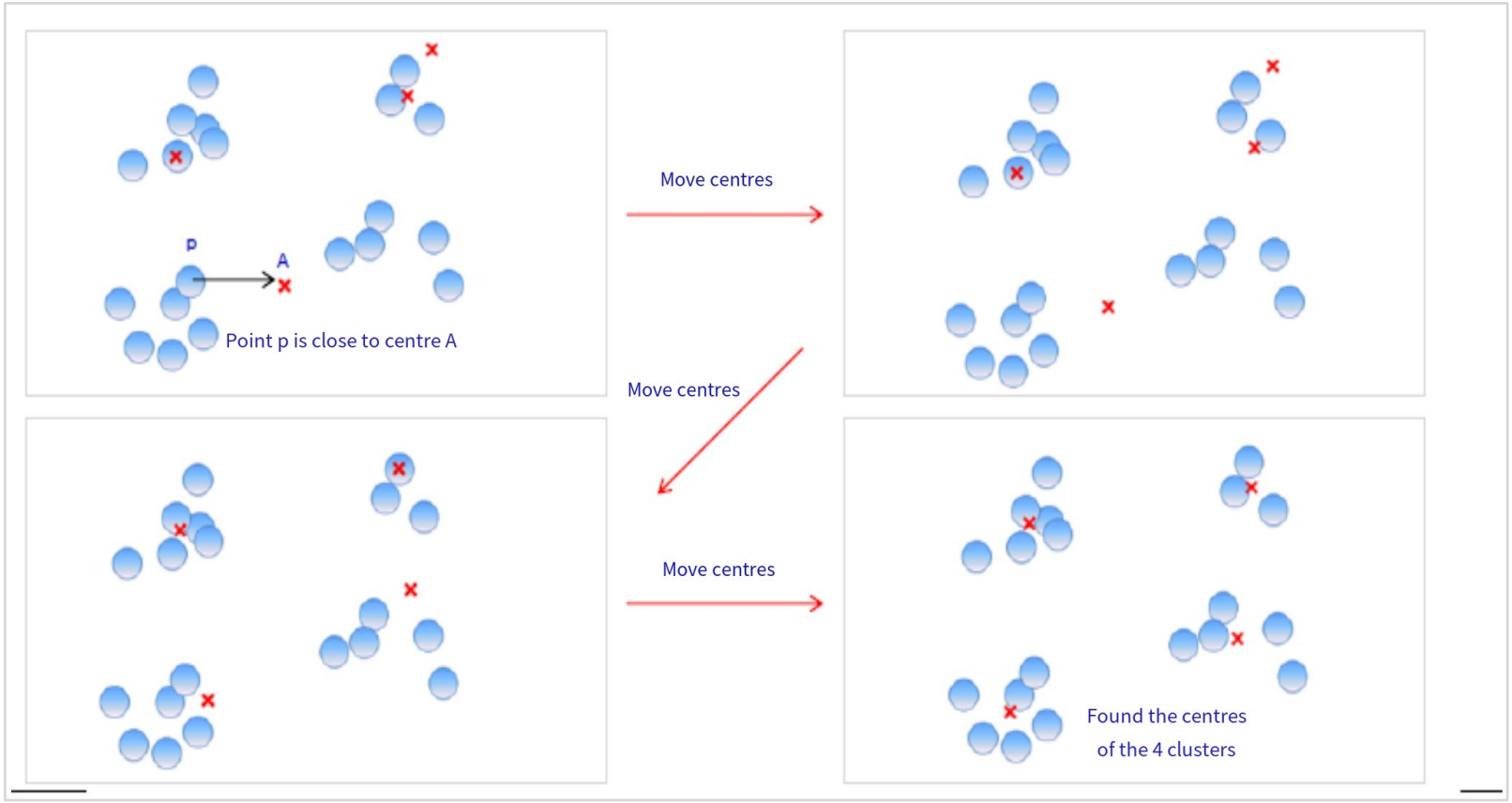
An unsupervised warm-up for hidden states

K-Means alternates between assigning points and moving centres

Unsupervised learning · an EM-type loop over centres and clusters

Loop

- Pick K centres (×)
- Assign each point to nearest
- Move centre to cluster mean
- Repeat until stable



Daily candles can be clustered by their open-high-low-close shape

K-Means clustering of candlesticks in a spreadsheet

Set-up

- Rows: daily OHLC
- Normalise each candle
- Assign to 4 clusters
- Recompute centres

Stock candlestick clustering (K-Means Clustering)

Training data

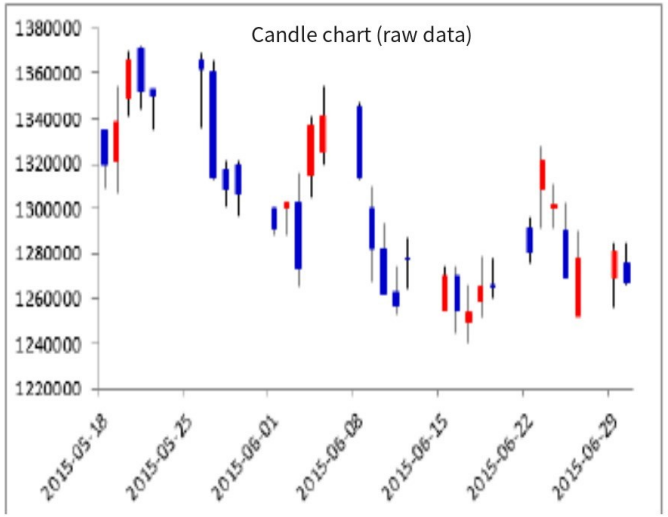
Open	High	Low	Close	Class
0.6341	0.6341	0.3859	0.4909	3
0.4483	0.7715	0.3152	0.6099	2
0.4663	0.6621	0.3918	0.6248	4
0.6408	0.6501	0.3898	0.4642	1
0.5855	0.5855	0.4166	0.5573	3
0.6084	0.6363	0.3291	0.5711	3
0.7417	0.7984	0.2977	0.3071	1
0.5844	0.6230	0.4302	0.5073	3
0.6206	0.6303	0.3988	0.4953	3
0.5851	0.5948	0.4679	0.4972	3
0.5508	0.5801	0.4340	0.5801	3
0.6711	0.7986	0.2984	0.3769	1
0.4455	0.6937	0.3501	0.6556	4
0.4392	0.7233	0.3918	0.5907	4
0.6812	0.7003	0.3770	0.3865	1
0.6343	0.7323	0.3206	0.4578	1
0.6057	0.7247	0.4073	0.4073	1
0.5488	0.6590	0.4486	0.4886	3
0.5511	0.6402	0.4124	0.5412	3
0.4512	0.6413	0.4512	0.6013	4
0.6265	0.6666	0.3758	0.4761	1
0.5110	0.6725	0.4100	0.5514	4
0.4912	0.6913	0.4111	0.5512	4
0.5213	0.6410	0.4614	0.5213	3

Cluster centres

	Open	High	Low	Close
1	0.907	0.320	0.048	0.401
2	0.420	0.060	0.237	0.857
3	0.229	0.356	0.434	0.264
4	0.265	0.724	0.150	0.898

Reset | Step

Error 19.5752



Training data to centre distance

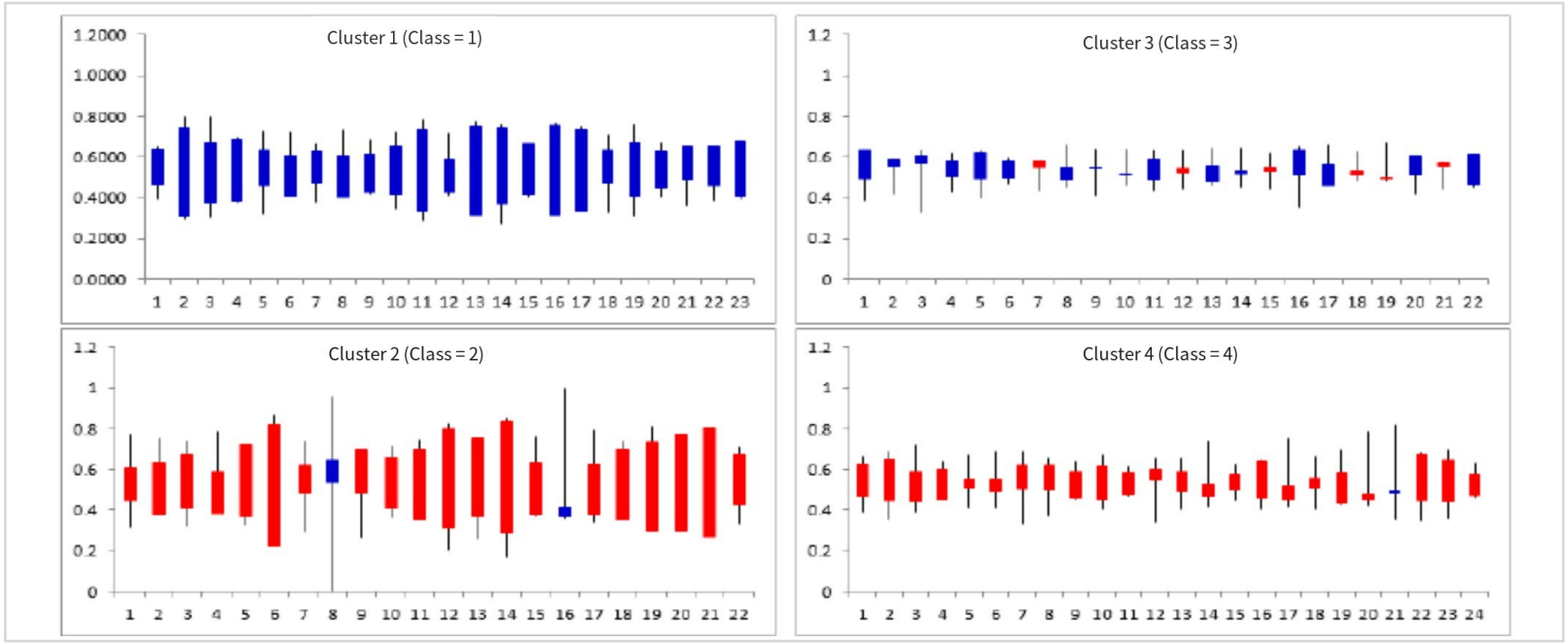
1	2	3	4	G1	G2	G3	G4	Dist.
0.1250	0.3635	0.1000	0.1994	0	0	1	0	0.1000
0.3135	0.1010	0.2386	0.1395	0	1	0	0	0.1010
0.3082	0.1893	0.1507	0.0449	0	0	0	1	0.0449
0.0965	0.3774	0.1165	0.2160	1	0	0	0	0.0965
0.2241	0.3363	0.0652	0.1522	0	0	1	0	0.0652
0.1951	0.2966	0.1333	0.1698	0	0	1	0	0.1333
0.1594	0.5191	0.3616	0.4271	1	0	0	0	0.1594
0.1776	0.3423	0.0328	0.1528	0	0	1	0	0.0328
0.1385	0.3572	0.0809	0.1855	0	0	1	0	0.0809
0.2053	0.3779	0.0548	0.1782	0	0	1	0	0.0548
0.2622	0.3192	0.0744	0.1294	0	0	1	0	0.0744
0.1094	0.4199	0.2906	0.3379	1	0	0	0	0.1094
0.3395	0.1299	0.2083	0.0965	0	0	0	1	0.0965
0.3017	0.1587	0.1754	0.0587	0	0	0	1	0.0587
0.0304	0.4371	0.2084	0.2970	1	0	0	0	0.0304
0.0782	0.3427	0.1899	0.2376	1	0	0	0	0.0782
0.0865	0.3766	0.1628	0.2326	1	0	0	0	0.0865
0.1838	0.3247	0.0486	0.1344	0	0	1	0	0.0486
0.2041	0.2902	0.0384	0.1025	0	0	1	0	0.0384
0.3192	0.2397	0.1339	0.0559	0	0	0	1	0.0559
0.0979	0.3514	0.1135	0.1983	1	0	0	0	0.0979
0.2286	0.2457	0.0790	0.0572	0	0	0	1	0.0572
0.2411	0.2278	0.1026	0.0463	0	0	0	1	0.0463
0.2295	0.3077	0.0442	0.1046	0	0	1	0	0.0442

Four clusters gather candles of similar shape

Each panel shows the candles of one cluster (class)

Result

- Class 1: large down candles
- Class 2: large up candles
- Class 3, 4: small bodies

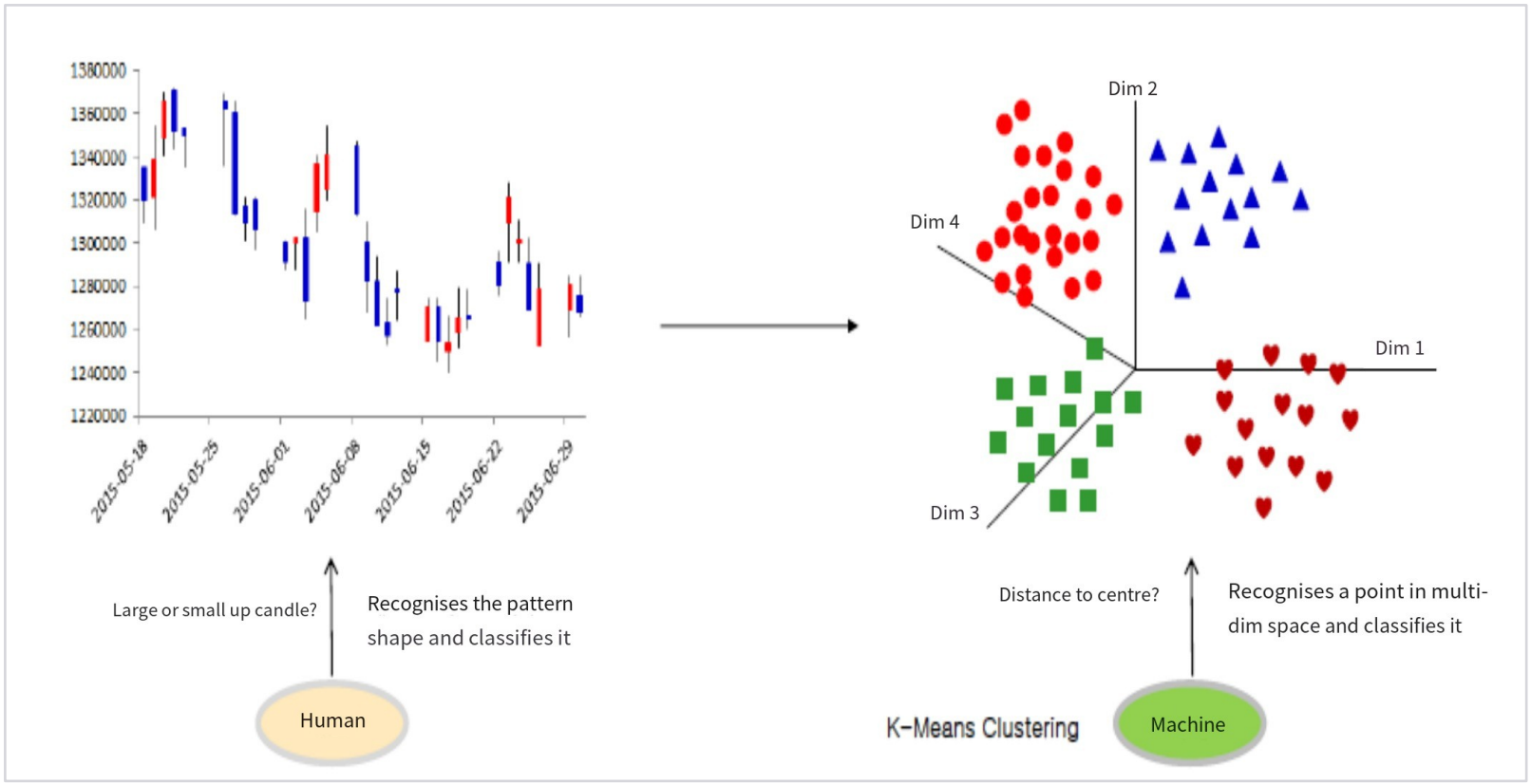


People read candle shapes by eye; K-Means does it by distance

Human and machine · 2–4 dimensional feature space

Human vs machine

- Human: up or down?
- Machine: distance in space
- Same idea, automated
- No time order yet



02

PART 2 · MARKOV CHAIN

Markov Chain and Hidden Markov Model

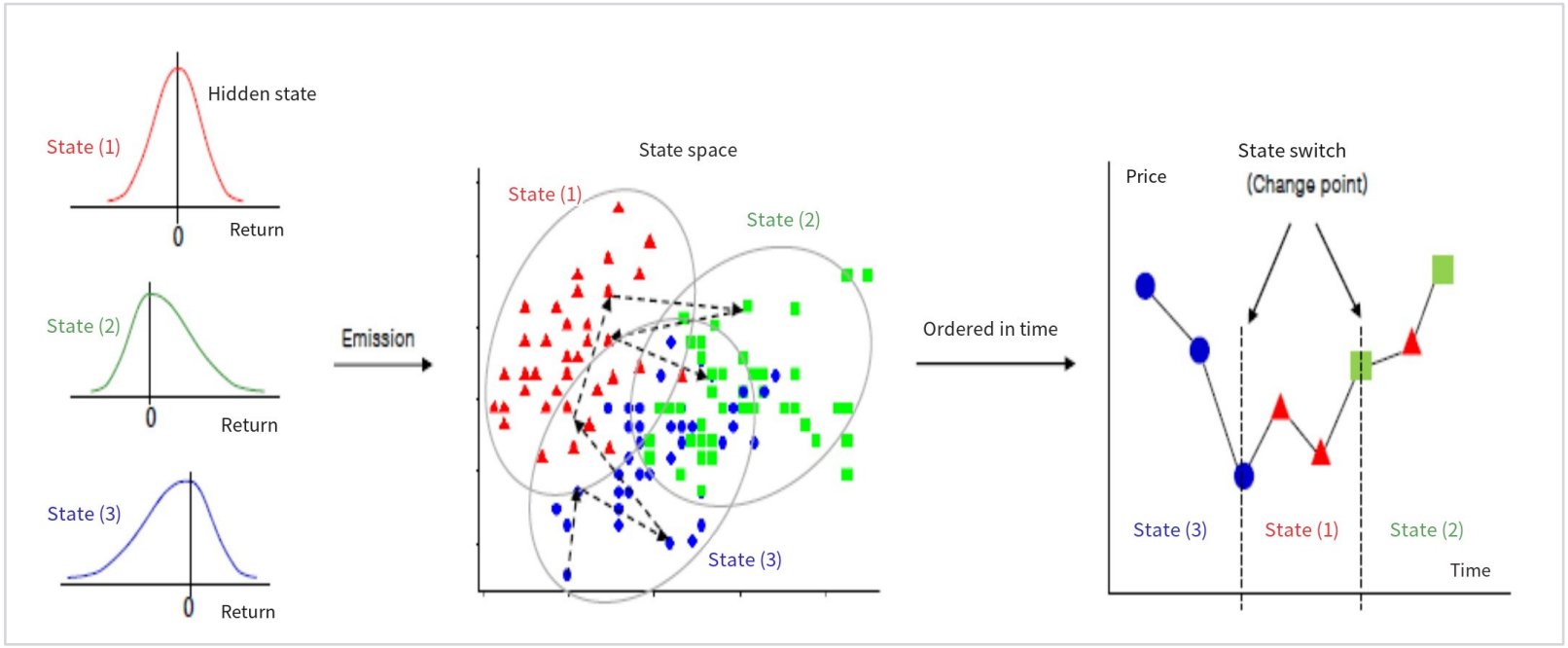
Transitions, emissions and the three tasks

Returns emitted from hidden states shift when the state changes

Emission → state space → change points over time

Picture

- Each state: a return distribution
- Points mix in the return space
- Time order reveals change points



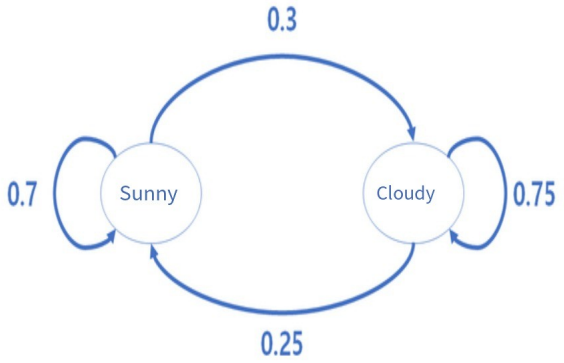
A transition matrix projects the weather two days ahead

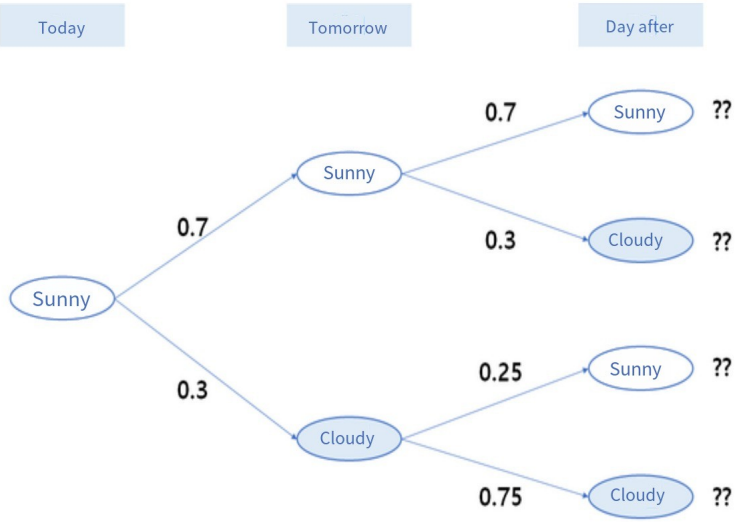
Sunny and cloudy · $P^2 = P \cdot P$ gives two-step probabilities

Two-step

- P = [[0.7, 0.3], [0.25, 0.75]]
- P^2 row 1: 0.565, 0.435
- P^2 row 2: 0.3625, 0.6375
- Tree shows the same sums

Today/Tomorrow	Sunny	Cloudy
Sunny	0.7	0.3
Cloudy	0.25	0.75





$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.25 & 0.75 \end{bmatrix}$$

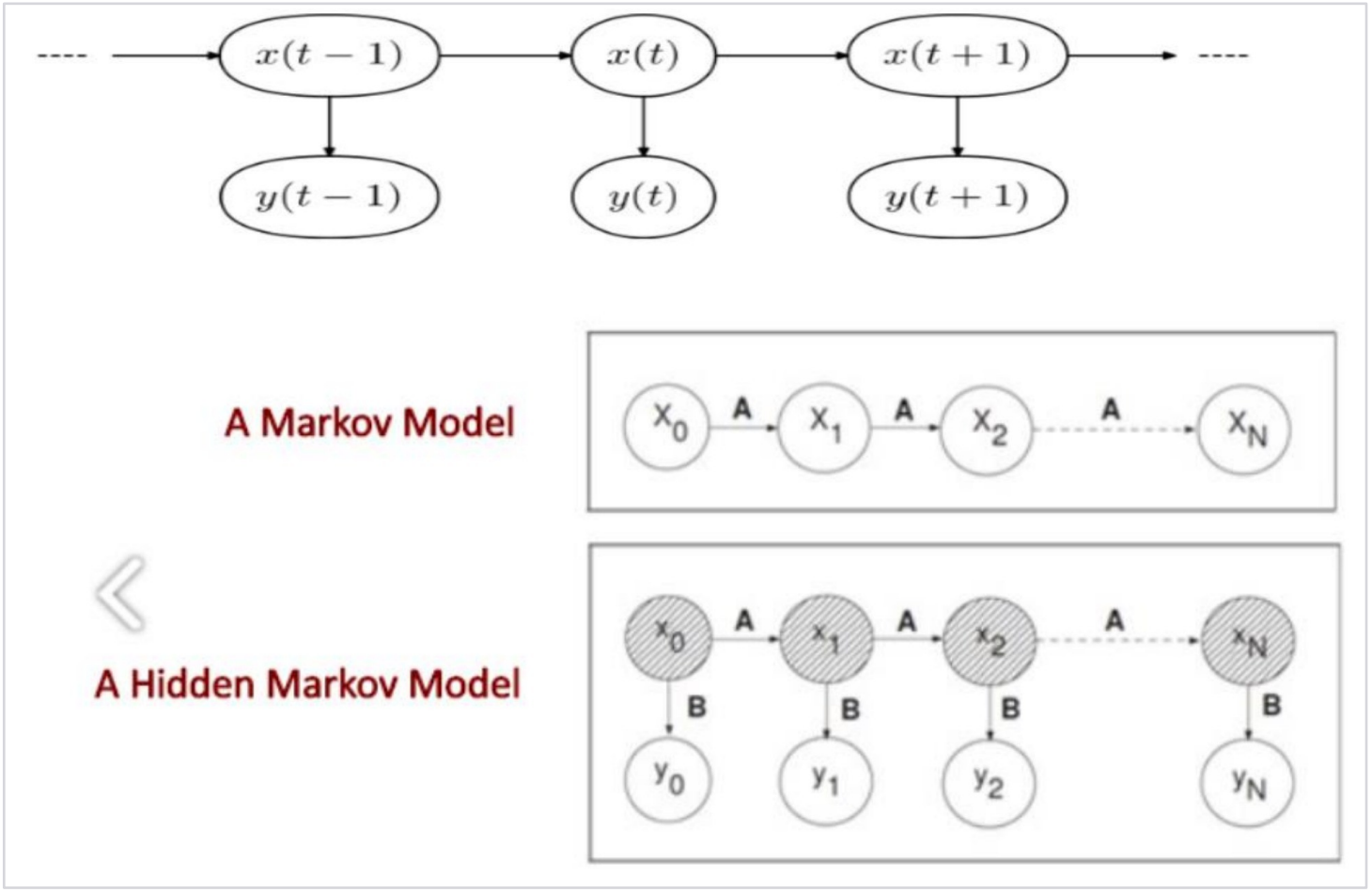
$$P2 = \begin{bmatrix} 0.7 & 0.3 \\ 0.25 & 0.75 \end{bmatrix} \cdot \begin{bmatrix} 0.7 & 0.3 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.565 & 0.435 \\ 0.3625 & 0.6375 \end{bmatrix}$$

An HMM attaches an observation to every state of a Markov chain

Markov model: x(t) only · HMM: x(t) hidden, y(t) observed

Compare

- Markov: A only
- HMM: A and B
- y depends on x of same time

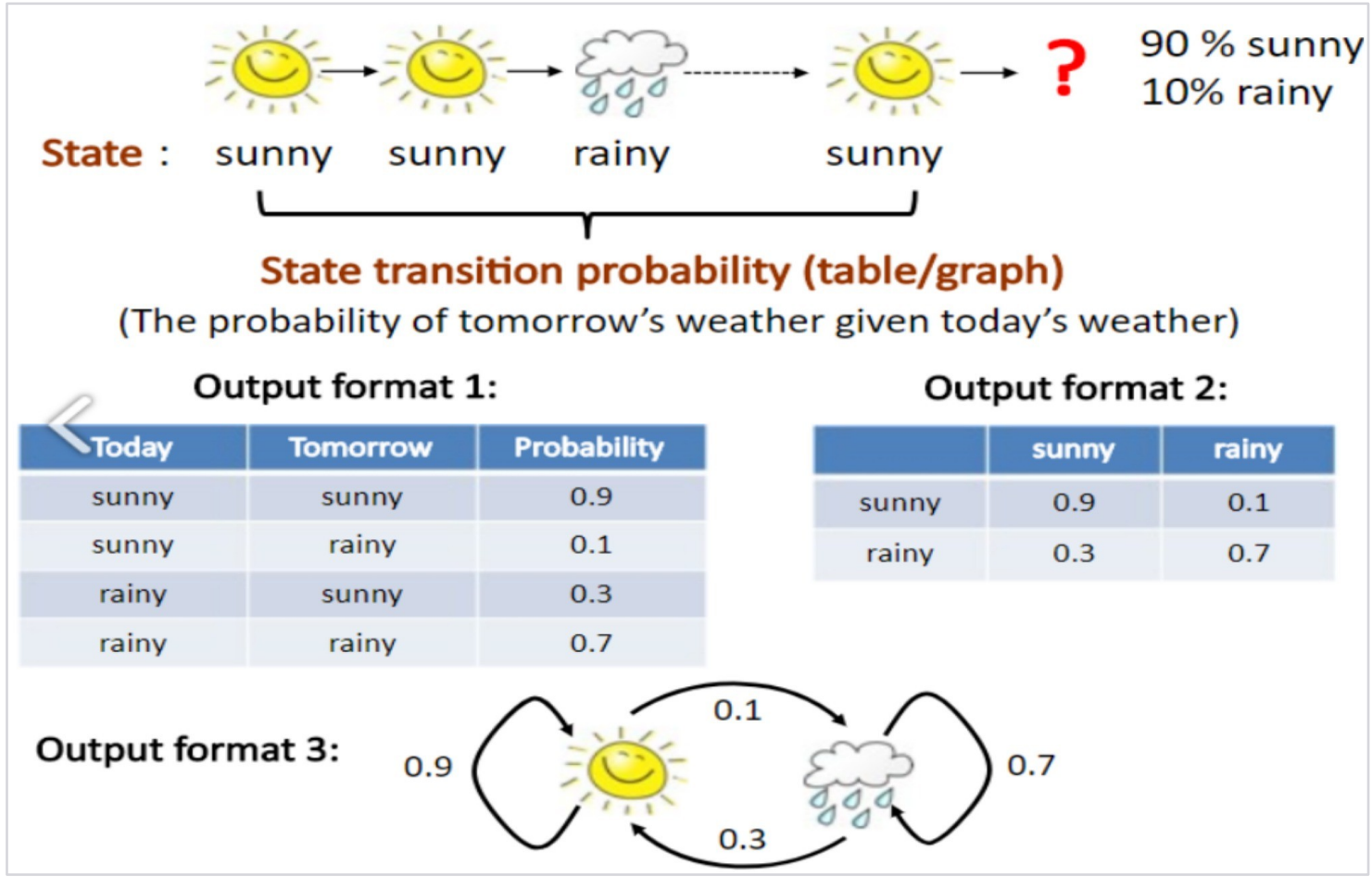


Two weather states give a 2×2 transition probability matrix

Probability of tomorrow's weather given today's weather

Three formats

- Table: today → tomorrow
- Matrix: rows sum to 1
- Graph: arrows and loops



Today	Tomorrow	Probability
sunny	sunny	0.9
sunny	rainy	0.1
rainy	sunny	0.3
rainy	rainy	0.7

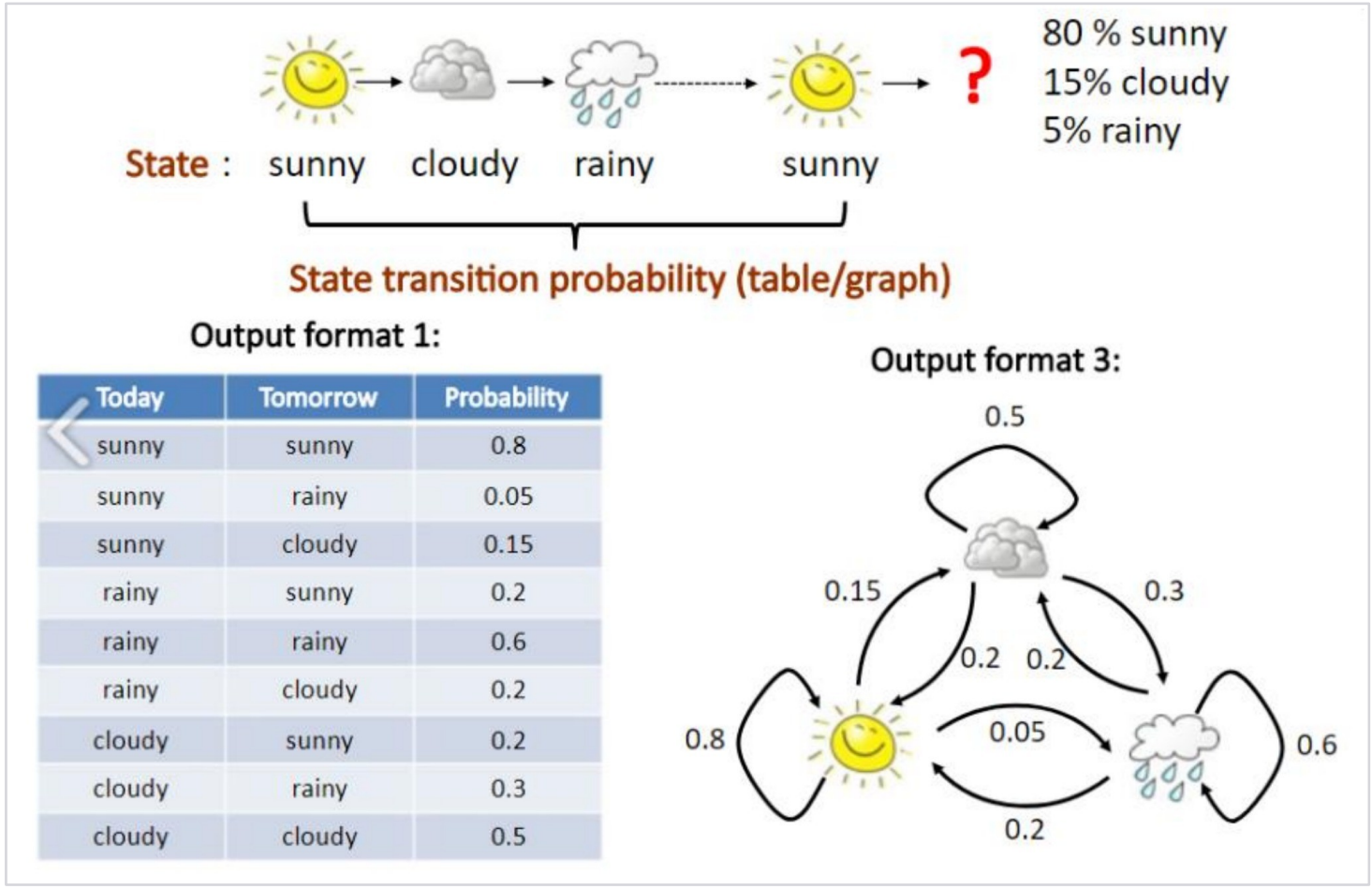
	sunny	rainy
sunny	0.9	0.1
rainy	0.3	0.7

Three weather states give a 3×3 transition matrix

Sunny, cloudy, rainy · output formats 1 and 3

Read

- Sunny→sunny 0.8
- Rainy→rainy 0.6
- Cloudy→rainy 0.3



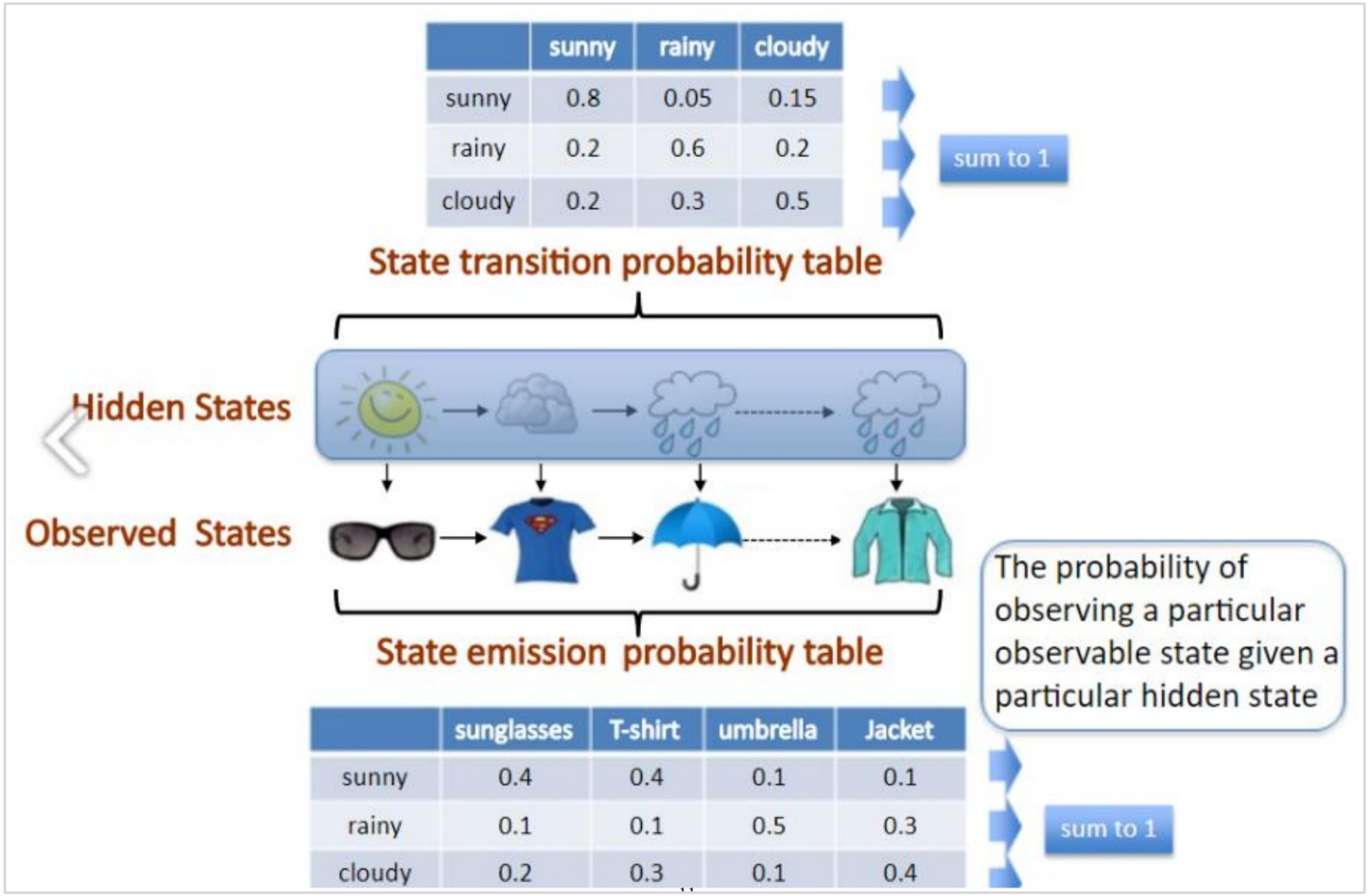
Today	Tomorrow	Probability
sunny	sunny	0.8
sunny	rainy	0.05
sunny	cloudy	0.15
rainy	sunny	0.2
rainy	rainy	0.6
rainy	cloudy	0.2
cloudy	sunny	0.2
cloudy	rainy	0.3
cloudy	cloudy	0.5

Emission probabilities link hidden weather to observed clothing

Hidden: weather · Observed: sunglasses, T-shirt, umbrella, jacket

Two tables

- Transition: state → state
- Emission: state → item
- Each row sums to 1



Each HMM task has its own algorithm

Initial, transition and emission probabilities define the model

Evaluation

Forward-Backward

Given the initial, transition and emission probabilities, compute the probability of the observations.

Decoding

Viterbi

Given the model and observations such as walk → clean → shop → shop → clean, find the most likely hidden sequence, e.g. sunny → rainy → sunny → rainy → rainy.

Learning

Baum-Welch

Given only observations, estimate the initial, transition and emission probabilities and the hidden sequence.

03

PART 3 · ALGORITHMS

Forward, Viterbi and Baum-Welch

Recursions and a worked example

The forward algorithm turns the joint probability into a recursion

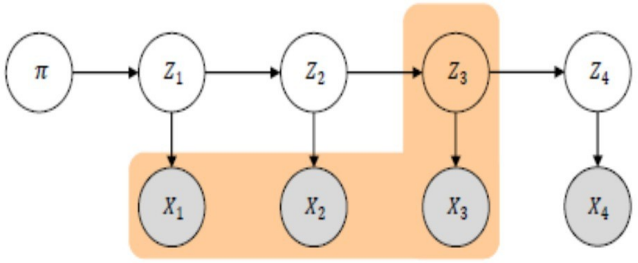
Sum over the previous state, then multiply by transition and emission

Derivation

- Split on z(k–1)
- Factor into three terms
- Recursion → dynamic programming
- Backward adds later x

- Forward Algorithm

$$P(z_k, x_{1:k}) = \begin{cases} z_{k-1} = Rainy \rightarrow P(z_k, z_{k-1} = Rainy, x_{1:k}) \\ z_{k-1} = Sunny \rightarrow P(z_k, z_{k-1} = Sunny, x_{1:k}) \end{cases} (+)$$



$$P(z_k, x_{1:k}) = \sum_{z_{k-1}} P(z_k, z_{k-1}, x_{1:k-1}, x_k)$$

The forward algorithm treats z3 as affected by x1, x2, x3. But z3 can also be affected by x4 and later observations. The algorithm that also uses observations from x4 on is the backward algorithm.

$$P(z_k, x_{1:k}) = \sum_{z_{k-1}} P(x_k|z_k, z_{k-1}, x_{1:k-1})\, P(z_k|z_{k-1}, x_{1:k-1}) P(z_{k-1}|x_{1:k-1}) P(x_{1:k-1})$$

$$P(z_k, x_{1:k}) = \sum_{z_{k-1}} P(x_k|z_k)\, P(z_k|z_{k-1}) P(z_{k-1}, x_{1:k-1}) \longrightarrow \alpha_k(z_k) = \sum_{z_{k-1}} P(x_k|z_k)\, P(z_k|z_{k-1}) \alpha_{k-1}(z_{k-1})$$

Recursion: a form solvable by dynamic programming

$$\alpha_1(z_1) = P(z_1, x_1) = P(x_1|z_1) P(z_1)$$

Walk, clean and shop have probability 3.10% under this HMM

π = [0.6, 0.4] · forward algorithm worked out in three steps

Result

- k=1: 0.06, 0.24
- k=2: 0.069, 0.0162
- k=3: 0.0219, 0.0091
- Sum = 0.031038

Forward algorithm exercise

- What is the probability of observing the sequence below under the HMM at right?
- We could compute every Z-sequence probability and add them all up, but we compute it recursively with the forward algorithm.

$X = Walk \rightarrow Clean \rightarrow Shop$

Initial prob. (π) = [0.6, 0.4]

Transition prob. (a)

	Rainy	Sunny
Rainy	0.7	0.3
Sunny	0.4	0.6

Emission prob. (b)

	Walk	Shop	Clean
Rainy	0.1	0.4	0.5
Sunny	0.6	0.3	0.1

$$\alpha_k(z_k) = \sum_{z_{k-1}} P(x_k|z_k)\, P(z_k|z_{k-1})\alpha_{k-1}(z_{k-1})$$

$$\alpha_1(z_1) = P(z_1, x_1) = P(x_1|z_1)P(z_1)$$

k = 1:

$$\alpha_1(z_1 = R) = P(z_1 = R, x_1 = W) = P(x_1 = W|z_1 = R)P(z_1 = R) = 0.1 * 0.6 = 0.06$$

$$\alpha_1(z_1 = S) = P(z_1 = S, x_1 = W) = P(x_1 = W|z_1 = S)P(z_1 = S) = 0.6 * 0.4 = 0.24$$

k = 2:

$$\alpha_2(z_2 = R) = P(z_2 = R, x_1 = W, x_2 = C) = \begin{cases} z_1 = R:\ 0.5 * 0.7 * 0.06 = 0.021 \\ z_1 = S:\ 0.5 * 0.4 * 0.24 = 0.048 \end{cases} \rightarrow \Sigma = 0.069$$

$$\alpha_2(z_2 = S) = P(z_2 = S, x_1 = W, x_2 = C) = \begin{cases} z_1 = R:\ 0.1 * 0.3 * 0.06 = 0.0018 \\ z_1 = S:\ 0.1 * 0.6 * 0.24 = 0.0144 \end{cases} \rightarrow \Sigma = 0.0162$$

k = 3:

$$\alpha_3(z_3 = R) = P(z_3 = R, x_1 = W, x_2 = C, x_3 = S) = \begin{cases} z_2 = R: 0.4 * 0.7 * 0.069 = 0.01932 \\ z_2 = S: 0.4 * 0.4 * 0.0162 = 0.002592 \end{cases} \rightarrow \Sigma = 0.021912$$

$$\alpha_3(z_3 = S) = P(z_3 = S, x_1 = W, x_2 = C, x_3 = S) = \begin{cases} z_2 = R: 0.3 * 0.3 * 0.069 = 0.00621 \\ z_2 = S: 0.3 * 0.6 * 0.0162 = 0.002916 \end{cases} \rightarrow \Sigma = 0.009126$$

$$\rightarrow \Sigma = 0.031038 = 3.1038\%$$

Viterbi replaces the forward sum with a maximum

Recursions from the original slides

$$\alpha_k(z_k) = \sum_{z_{k-1}} P(x_k \mid z_k)\, P(z_k \mid z_{k-1})\, \alpha_{k-1}(z_{k-1}), \qquad \alpha_1(z_1) = P(x_1 \mid z_1) P(z_1)$$

$$\mu_k(z_k) = \max_{z_{k-1}} P(x_k \mid z_k)\, P(z_k \mid z_{k-1})\, \mu_{k-1}(z_{k-1}), \qquad \mu_1(z_1) = P(x_1 \mid z_1) P(z_1)$$

Compare

- Forward: Σ over z(k−1) → probability of the observations
- Viterbi: max over z(k−1) → most likely hidden sequence

The Viterbi algorithm keeps the best path into each state

Derivation uses max(f·g) = f·max g when f ≥ 0

Derivation

- Same factorisation
- Σ becomes max
- Store the arg max
- Trace back at the end

▪Given π, a, b, estimate the most likely sequence Z (Z*) that could produce the observed X.

▪Build a recursive algorithm, as with the forward algorithm.

$$Z = \{z_1, z_2, z_3, \cdots\}, \qquad X = \{x_1, x_2, x_3, \cdots\}$$

Goal: $Z^* = \underset{Z}{\operatorname{argmax}} P(Z|X) = \underset{Z}{\operatorname{argmax}} P(Z, X)$

Def.: $\mu_k(z_k) = \max_{z_{1:k-1}} P(z_{1:k}, x_{1:k})$

$$\mu_k(z_k) = \max_{z_{1:k-1}} P(x_k|z_k)\, P(z_k|z_{k-1}) P(z_{1:k-1}, x_{1:k-1})$$

$$\mu_k(z_k) = \max_{z_{k-1}} \left[P(x_k|z_k) P(z_k|z_{k-1}) \max_{z_{1:k-2}} P(z_{1:k-1}, x_{1:k-1}) \right]$$

$$\mu_k(z_k) = \max_{z_{k-1}} P(x_k|z_k) P(z_k|z_{k-1}) \mu_{k-1}(z_{k-1})$$

Recursion: a form solvable by dynamic programming

Theorem (1)

If $f(a) \geq 0, g(a,b) \geq 0$, *then*

$$\max_{a,b} f(a) g(a,b) = \max_a \left[f(a) \max_b g(a,b) \right]$$

Viterbi algorithm

$$\mu_k(z_k) = \max_{z_{k-1}} P(x_k|z_k) P(z_k|z_{k-1}) \mu_{k-1}(z_{k-1})$$

$$\mu_1(z_1) = P(z_1, x_1) = P(x_1|z_1) P(z_1)$$

Sunny, rainy, rainy, sunny best explains walk, clean, shop, walk

Most likely hidden sequence has probability 0.2419%

Result

- k=4: R 0.000941, S 0.002419
- End at sunny
- Trace back via red arrows
- Z = S → R → R → S

Viterbi algorithm exercise

- Given the sequence below, what is the most likely Z-sequence?
- We could compute every Z-sequence probability and take the largest, but we compute it recursively with the Viterbi algorithm.

$$X = Walk \rightarrow Clean \rightarrow Shop \rightarrow Walk$$

$$\mu_k(z_k) = \max_{z_{k-1}} P(x_k|z_k)P(z_k|z_{k-1})\mu_{k-1}(z_{k-1})$$

$$\mu_1(z_1) = P(z_1, x_1) = P(x_1|z_1)P(z_1)$$

$k = 1: x_1 = Walk$

$$\mu_1(z_1 = R) = P(x_1 = W|z_1 = R)P(z_1 = R) = 0.1 * 0.6 = 0.06$$
$$\mu_1(z_1 = S) = P(x_1 = W|z_1 = S)P(z_1 = S) = 0.6 * 0.4 = 0.24$$

$k = 2: x_2 = Clean$

$$\mu_2(z_2 = R) = max\begin{cases} z_1 = R: 0.5 * 0.7 * 0.06 = 0.021 \\ z_1 = S: 0.5 * 0.4 * 0.24 = 0.048 \end{cases}$$
$$\mu_2(z_2 = S) = max\begin{cases} z_1 = R: 0.1 * 0.3 * 0.06 = 0.0018 \\ z_1 = S: 0.1 * 0.6 * 0.24 = 0.0144 \end{cases}$$

$k = 3: x_3 = Shop$

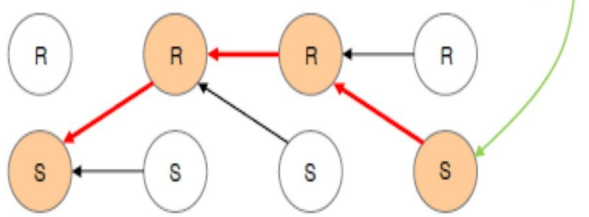
$$\mu_3(z_3 = R) = max\begin{cases} z_2 = R: 0.4 * 0.7 * 0.048 = 0.01344 \\ z_2 = S: 0.4 * 0.4 * 0.0144 = 0.002304 \end{cases}$$
$$\mu_3(z_3 = S) = max\begin{cases} z_2 = R: 0.3 * 0.3 * 0.048 = 0.00432 \\ z_2 = S: 0.3 * 0.6 * 0.0144 = 0.002592 \end{cases}$$

$k = 4: x_4 = Walk$

$$\mu_4(z_4 = R) = max\begin{cases} z_3 = R: 0.1 * 0.7 * 0.01344 = 0.000941 \\ z_3 = S: 0.1 * 0.4 * 0.00432 = 0.000173 \end{cases}$$
$$\mu_4(z_4 = S) = max\begin{cases} z_3 = R: 0.6 * 0.3 * 0.01344 = 0.002419 \\ z_3 = S: 0.6 * 0.6 * 0.00432 = 0.001555 \end{cases}$$

Trace back

Larger



$$Z = Sunny \rightarrow Rainy \rightarrow Rainy \rightarrow Sunny$$

Z* is this sequence, with the highest probability, 0.2419%.

Baum-Welch updates π, a and b by expectation-maximisation

E-step: Q(θ, θ_t) · M-step: θ_(t+1) = arg max Q under sum-to-one constraints

Steps

- Start from guessed θ
- E: expected log-likelihood
- M: maximise with Lagrange
- Repeat until stable

- Using only the observed X, estimate the most likely π, a, b, Z.
- Set initial θ = (π, a, b) at random, then update θ by repeating the E-M algorithm (Expectation and Maximization) on the observed X.

* E-step (Expectation): $Q(\theta,\theta_t)=\sum_Z P(Z|X,\theta_t)logP(X,Z|\theta)$

$\theta=\{\pi,a,b\}, Z=\{z_1,z_2,z_3,\cdots\}, X=\{x_1,x_2,x_3,\cdots\}$

* M-step (Maximization): $\theta_{t+1}=\underset{\theta}{\operatorname{argmax}}\, Q(\theta,\theta_t)$

- Parameter update

$$\pi_{t+1}=\frac{P(z_1=j|X,\theta_t)}{\sum_{j=1}^{k}P(z_1=j|X,\theta_t)}$$

$$a_{t+1}(i,j)=\frac{\sum_{t=2}^{T}P(z_{t-1}=i,z_t=j|X,\theta_t)}{\sum_{t=2}^{T}P(z_{t-1}=i|X,\theta_t)}$$

$$b_{t+1}(i,j)=\frac{\sum_{t=1}^{T}P(z_t=i|X,\theta_t)I(x_t=j)}{\sum_{t=1}^{T}P(z_t=i|X,\theta_t)}$$

- How the update rule is derived

1. Define Q(θ, θt) with the formula below (Factorization).

$$P(X,Z|\theta)=\pi\prod_{t=2}^{T}a(z_{t-1},z_t)\prod_{t=1}^{T}b(x_t,z_t)$$

$$logP(X,Z|\theta)=\log(\pi)+\sum_{t=2}^{T}\log\big(a(z_{t-1},z_t)\big)+\sum_{t=1}^{T}\log(b(x_t,z_t))$$

2. Write Q(θ, θt) as a Lagrangian using the constraints (Σπ = 1, Σa = 1, Σb = 1).

$$L(\theta,\theta_t)=Q(\theta,\theta_t)-\lambda_\pi\left(\sum_{i=1}^{k}\pi_i-1\right)-\lambda_a\big(a-\overline{\text{cond.}}\big)-\lambda_b\big(b-\overline{\text{cond.}}\big)$$

3. Find the parameters that maximise the Lagrangian.

$$\frac{\partial L}{\partial \pi_i}=0\quad \frac{\partial L}{\partial a_i}=0\quad \frac{\partial L}{\partial b_i}=0\quad \frac{\partial L}{\partial \lambda_\pi}=0\quad \frac{\partial L}{\partial \lambda_a}=0\quad \frac{\partial L}{\partial \lambda_b}=0$$

04

PART 4 · STOCK PREDICTION

Stock Price Prediction with HMM

Model selection, price prediction and return regimes on AAPL and KOSPI

Information criteria choose the number of hidden states

Gaussian HMM, N states, M observations, parameters fitted by Baum-Welch

Read the page

- L: likelihood
- M: number of observations
- k: number of parameters
- Paper: states 2–4 only

Choosing the number of hidden states in a Hidden Markov Model (HMM) is an important task. We usually assess model performance with AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion), which suit maximum likelihood estimation with the Baum-Welch algorithm. This study limits the number of states to 2–4 to keep the model simple and practical.

AIC and BIC are computed as follows:

$$AIC = -2\ln(L) + 2k$$

$$BIC = -2\ln(L) + k\ln(M)$$

where:

- L is the model likelihood function,
- M is the number of observation points,
- k is the number of estimated parameters.

Each hidden state is assumed to follow a normal distribution. With N states in the HMM, the number of parameters k is defined as:

$$k = N^2 + 2N - 1$$

We run 100 simulations. At each step the parameters estimated in the previous step are the initial values for re-estimation, and AIC and BIC are computed. This shows how BIC changes across the simulations.

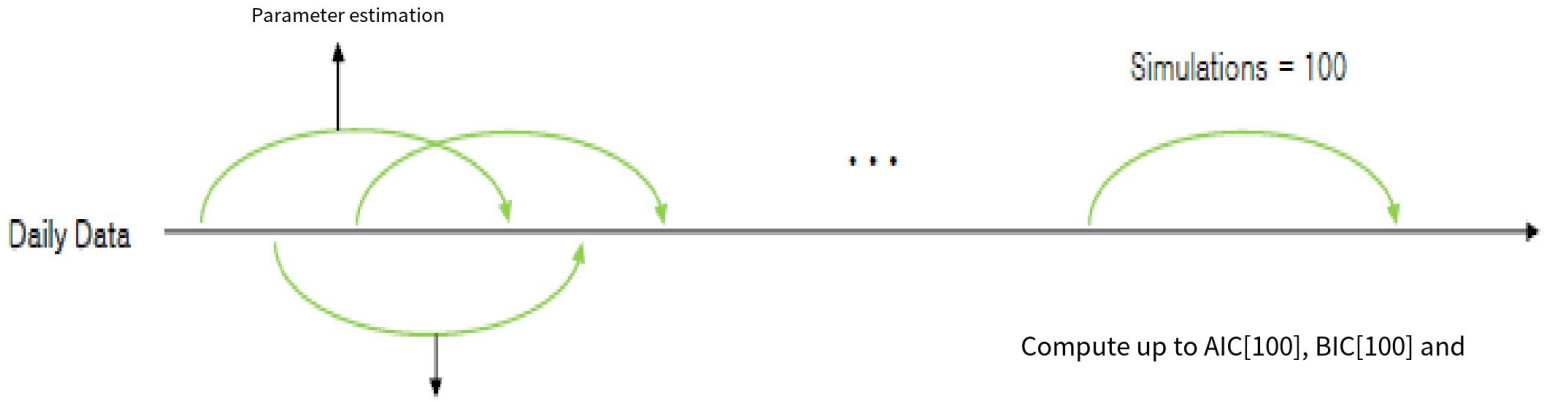
$$AIC = -2\ln(L) + 2k, \qquad BIC = -2\ln(L) + k\ln(M), \qquad k = N^2 + 2N - 1$$

Repeated calibration tracks AIC and BIC over 100 simulations

Each step starts from the previous step's parameters

Calibration loop

- Slide one step forward
- Re-estimate from last λ
- Record AIC[i], BIC[i]
- Repeat 100 times



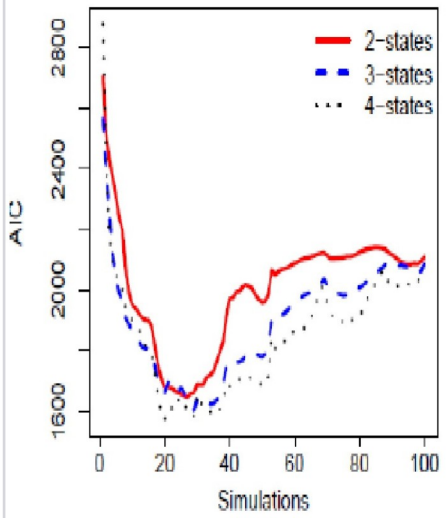
In the AAPL paper four states score lowest, and the gap persists on BIC

Apple daily data, 2015.1.6–2015.10.30 · left AIC, right BIC

Paper result

- AIC: 4 states lowest, gap narrows
- BIC: 4 states stay lowest
- Paper picks 4 states
- Not always U-shaped

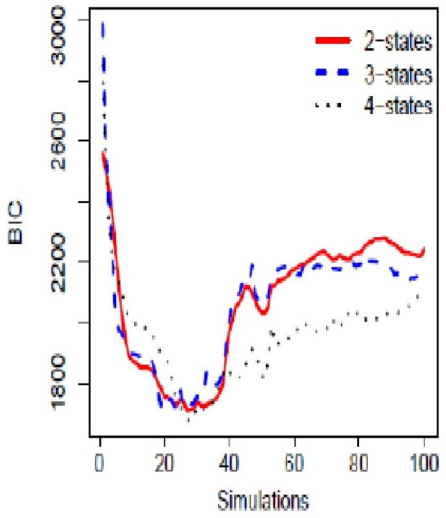
- Vary the number of states over 2, 3, 4 and measure AIC and BIC as on the previous page.
- The figure below is [Figure 1] of the paper: AIC and BIC by number of states for Apple.
- For AIC, 4 states give the lowest value, but as simulations increase AIC converges for all state counts, so the gap is small. For BIC, however, 4 states stay lowest even as simulations increase. So 4 states can be considered suitable for Apple (per the paper).
- AIC and BIC do not always show this U-shape. The shape can differ by stock and period.



Stock: Apple (AAPL) daily data

Period: 2015.1.6 – 2015.10.30

Figure 1: AIC (left) and BIC (right) for 100 parameter calibrations of HMM using Apple, AAPL, stock daily trading prices

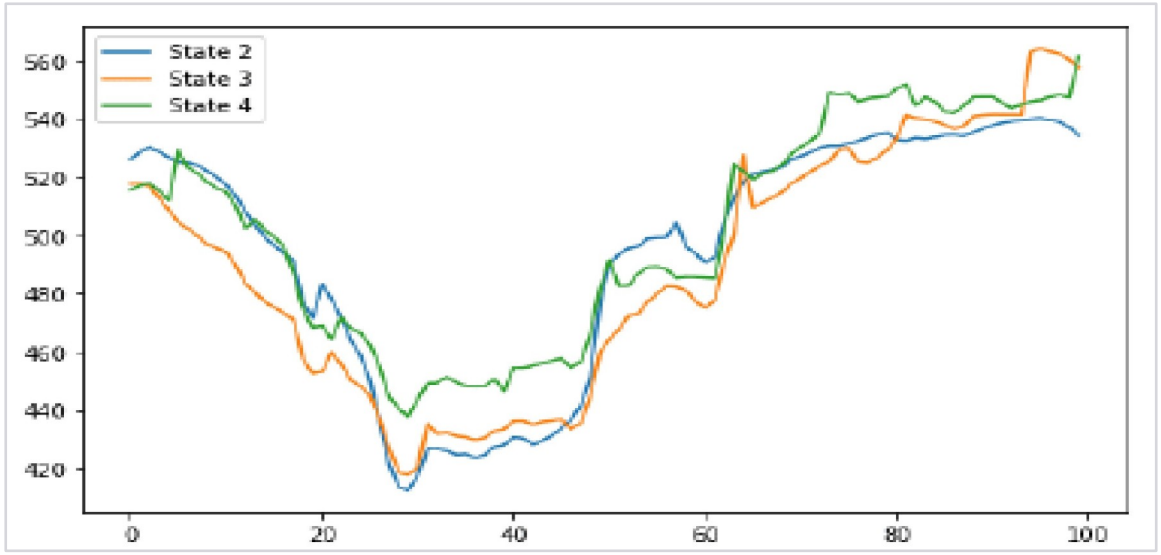


The BIC comparison is repeated on AAPL closing prices alone

hmmlearn, same period as the paper, 2–4 states

[Input] Prompt · Claude Code

```
In the hmm-bic folder, please do the following.
1. Get AAPL daily closing prices.
   Period: 2015.1.6–2015.10.30
2. Fit a hmmlearn Gaussian HMM with 2, 3 and
   4 states.
3. Run 100 simulations: slide one step forward
   and re-estimate from the last step's
   parameters. Propose the window length first.
4. Record BIC = −2 ln(L) + k ln(M) each time,
   with k = N² + 2N − 1.
5. Save the BIC curves per state count to
   bic_close.png and the values to bic_close.csv.
```



[Check] What a person reviews

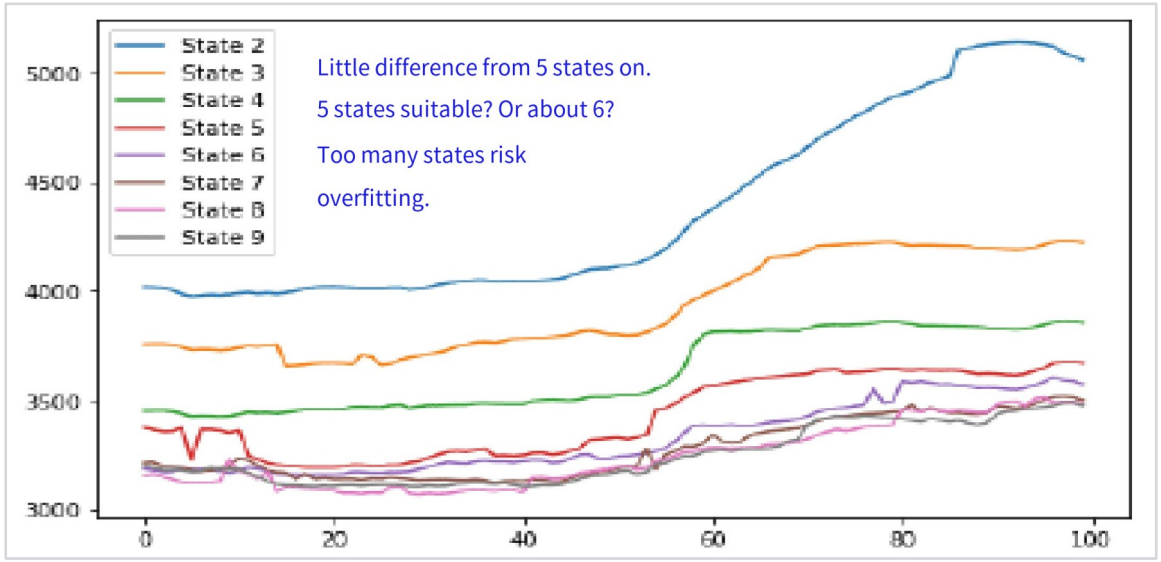
- Is $k = N^2 + 2N - 1$?
- Does each step start from the last parameters?
- Lower BIC is better: which state count is lowest?

With OHLC inputs, gains in BIC fade beyond about five states

Open, high, low, close · 2–9 states · too many states risk overfitting

[Input] Prompt · Claude Code

```
Continue in the hmm-bic folder.
1. Change the input to open, high, low and close.
   Keep the stock and period of the last slide.
2. Extend the states to 2–9.
3. Keep the last slide's 100 simulations and BIC
   steps; recount k for the input dimension.
4. Save the BIC curves per state count to
   bic_ohlc.png and the values to bic_ohlc.csv.
```



[Check] What a person reviews

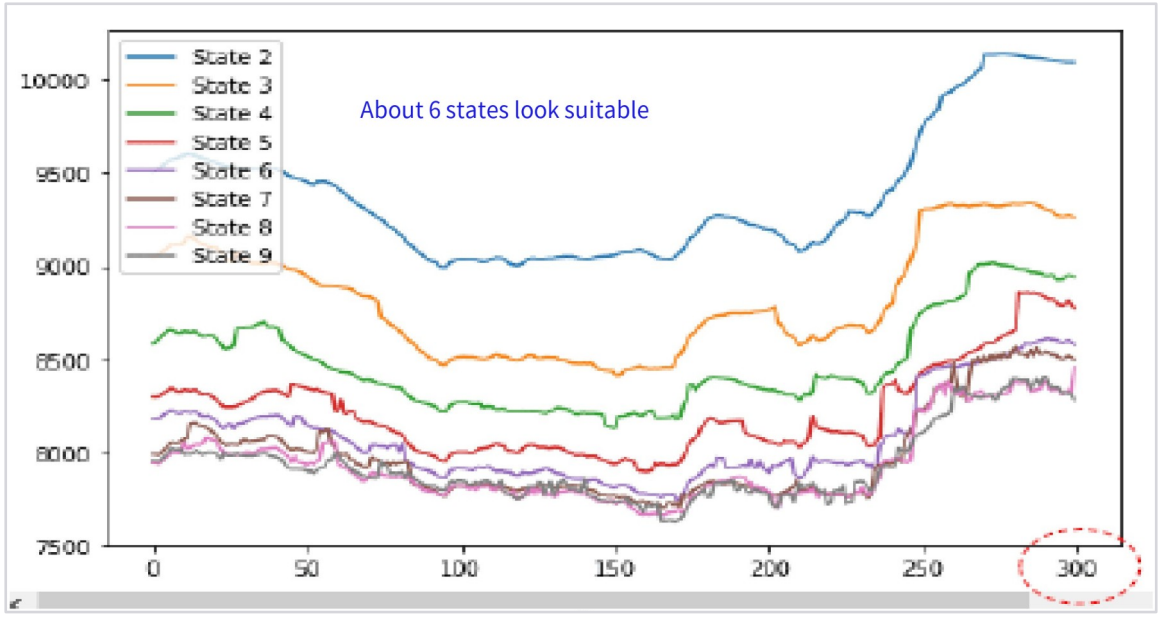
- Is k counted for the input dimension?
- Is the gain small after 5–6 states?
- Large N overfits: is the chosen N too large?

KOSPI index data point to six or seven hidden states

Slide text says 7; the chart note says 6 — curves for 6–9 states nearly overlap

[Input] Prompt · Claude Code

```
Continue in the hmm-bic folder.
1. Change the data to daily KOSPI composite
   index values and state the period used.
2. Keep the states at 2–9.
3. Run 300 simulations with the steps of the
   last slide.
4. Save the BIC curves per state count to
   bic_kospi.png and the values to bic_kospi.csv.
```



[Check] What a person reviews

- Is the period stated?
- Do the curves converge at 6–7 states?
- Where curves overlap, is the smaller N chosen?

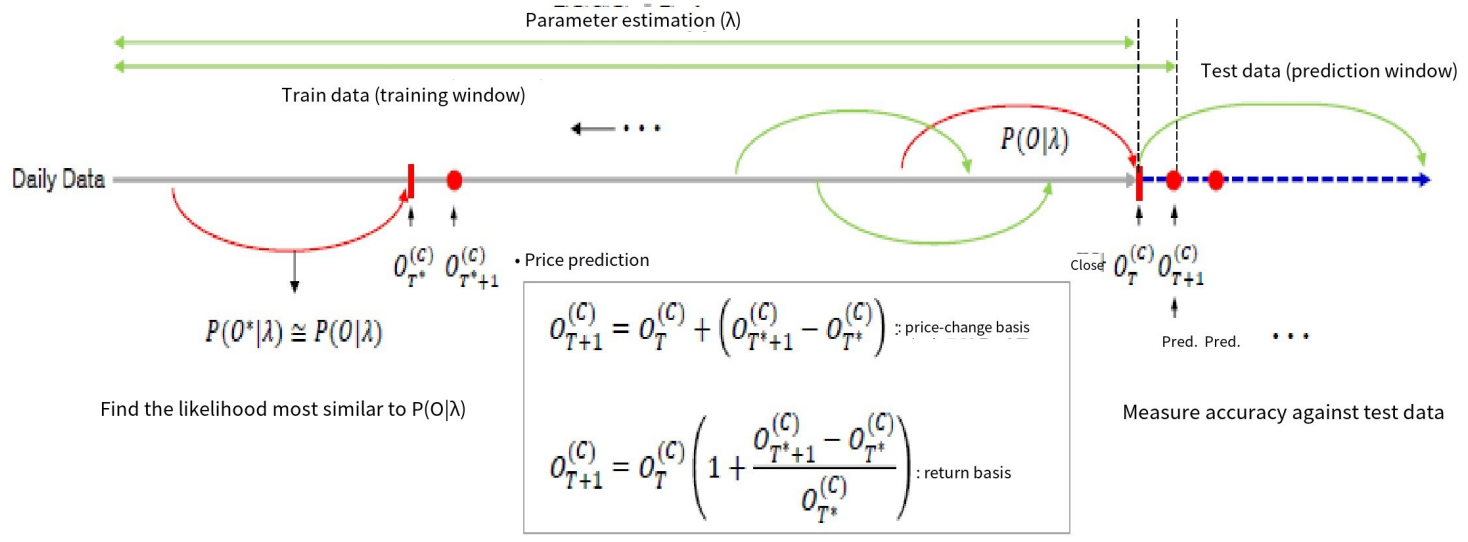
Prediction borrows the price change of the most similar past day

Train: fit λ · find day with likelihood closest to today · apply its change

Three steps

- 1 Calibrate λ, compute P(O|λ)
- 2 Find past day with similar P
- 3 Apply its change to today
- Two variants: price change or return

▪ We first introduce how to predict stock prices using HMM. The prediction process can be divided by three steps. Step one: calibrate HMM's parameters and calculate likelihood of the model. Step 2: find a day in the past that has a similar likelihood to that of the recent day. Finally, step 3: use the difference of stock prices on the "similar" day in the past to predict future stock prices. This prediction approach is based on the work of Hassan and Nath [2]. ~~However, our procedure is different from their method in that we put the sign of the difference of the likelihood of recent day with likelihood of the "similar" day in the past into the calculation of future stock prices~~. Furthermore, we do not only use HMM with multiple observations (open, low, high, close price), as in [2], but also the HMM with one observation data (close price) to predict future close price. By using both single and multiple observations to predict stocks' close prices, we can examine which method performed better. The HMM with a single observation is a special case of HMM with multiple observations, thus we will explain in detail how to use HMM with multiple observations.

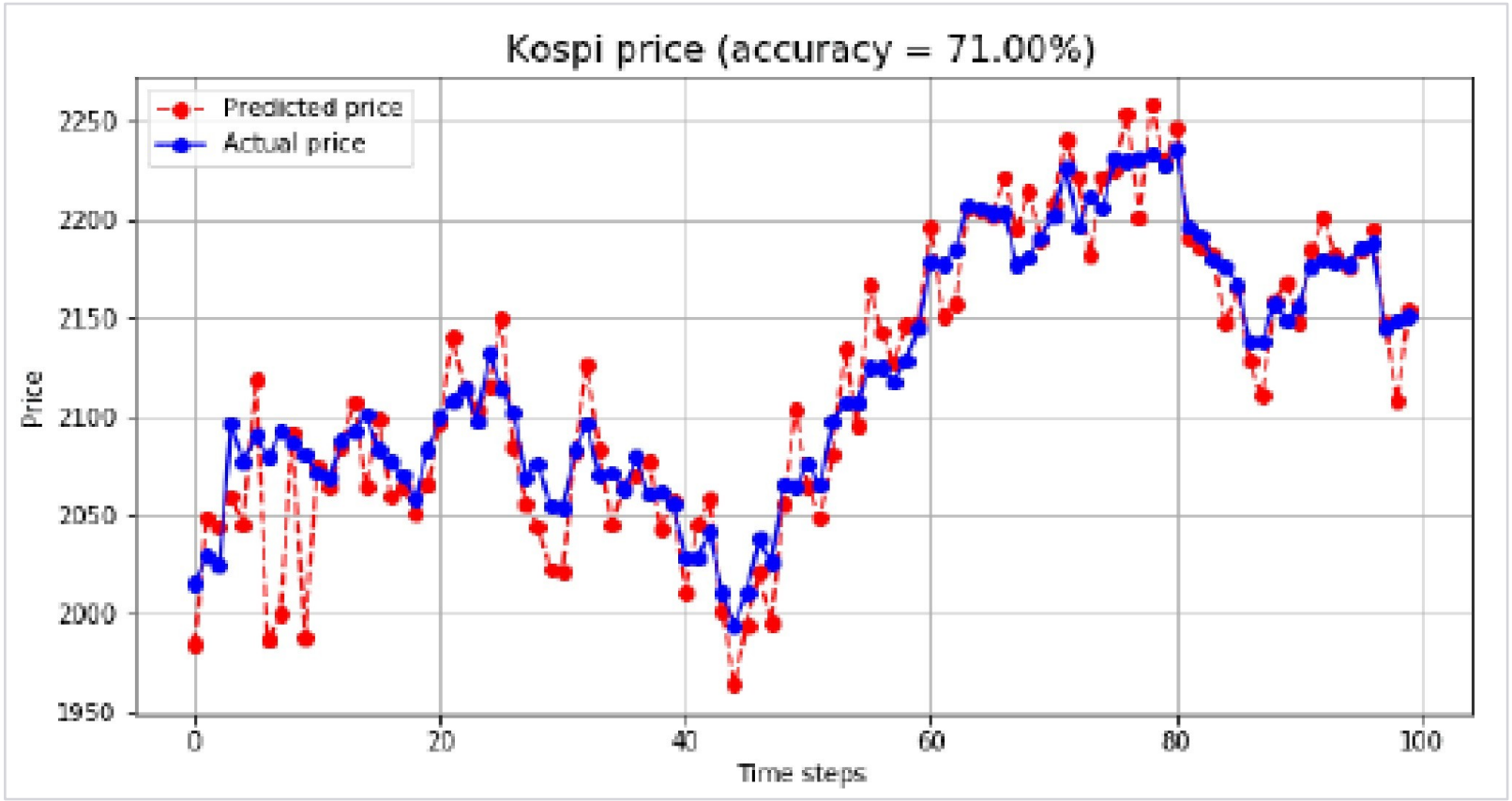


The rolling forecast tracks KOSPI over the last 100 days

Reported accuracy 71.00% on the test window

Result

- Red: predicted price
- Blue: actual price
- Accuracy = 71.00%
- Same logic → tomorrow



A three-state HMM splits KOSPI returns into distinct regimes

Estimated mean, volatility and transition matrix of each state

Estimates

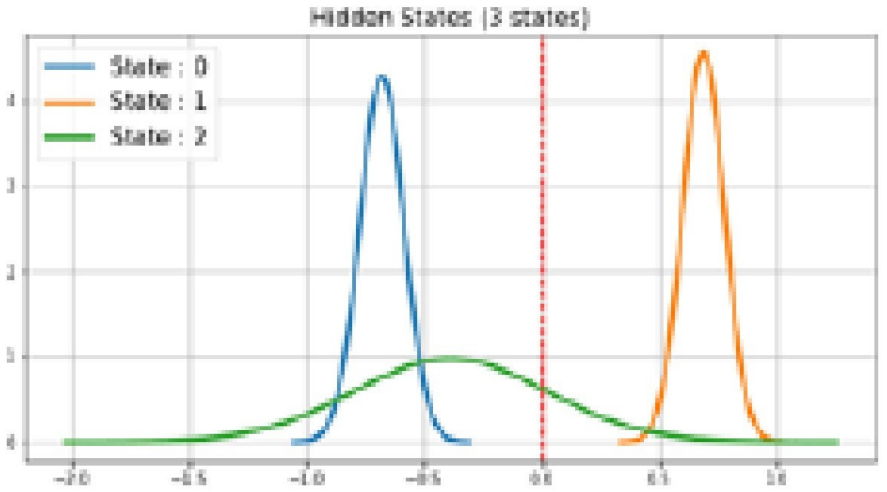
- μ: −0.681, 0.688, −0.389
- σ: 0.093, 0.088, 0.411
- Diagonal: 0.855, 0.90, 0.932
- Regimes are persistent

```
mu :
 [[-0.681]
 [ 0.688]
 [-0.389]]

vol :
 [[0.093]
 [0.088]
 [0.411]]

Transition :
 [[0.855 0.118 0.027]
 [0.099 0.9   0.002]
 [0.017 0.051 0.932]]
```

Mean and std dev of the 3 return distributions



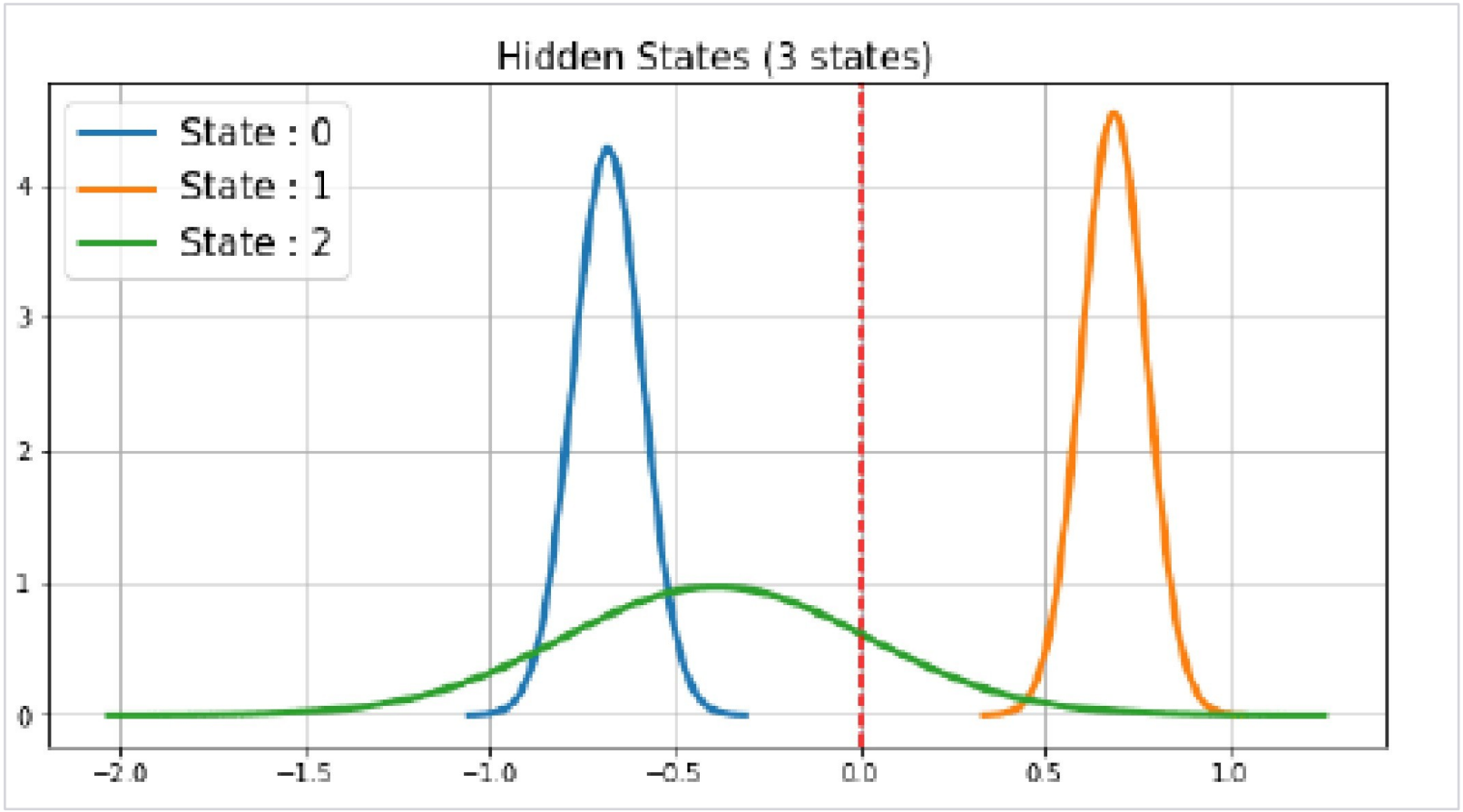
Transition probabilities of the 3 return distributions

The return distributions show two calm states and one volatile state

State 0: negative mean · State 1: positive mean · State 2: wide spread

Read the curves

- Blue: calm down market
- Orange: calm up market
- Green: high volatility
- Red line: zero return



The KOSPI closing price of the last 300 days is the reference picture

2018-01 to 2019-03 daily close

Look first

- Peak in early 2018
- Fall into late 2018
- Partial rebound in 2019

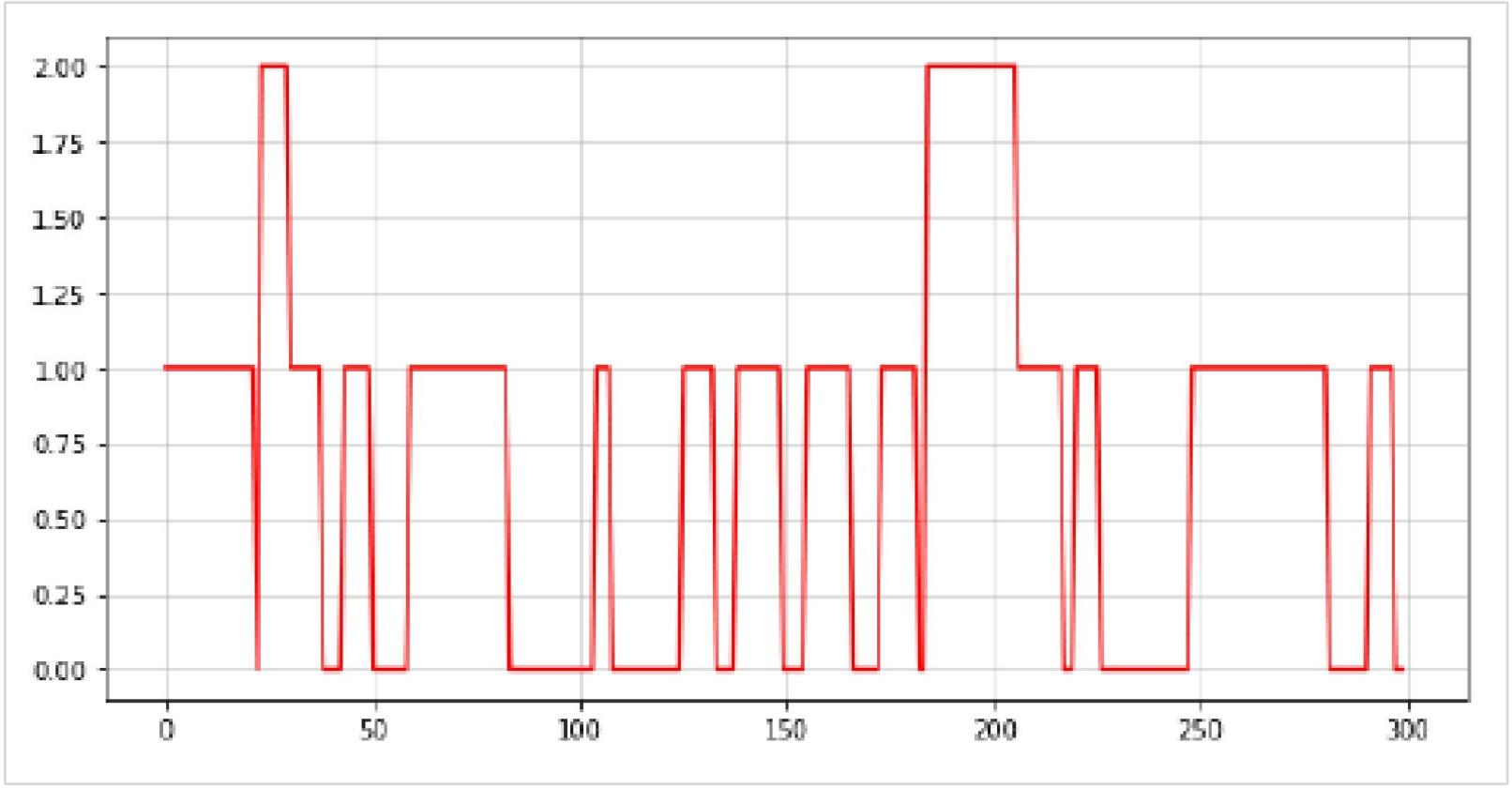


The decoded state path marks when the market switched regime

Last 300 days · current state 0 (mean −0.681, std 0.093 as stated in the slide)

Read the path

- y-axis: state 0, 1, 2
- Spikes to 2: stress
- Now: state 0



Parameters, distributions and the state path tell one story

Summary of the KOSPI return-regime exercise

Put together

- Table: μ, σ, A
- Curves: per-state returns
- Top right: state path
- Bottom right: price

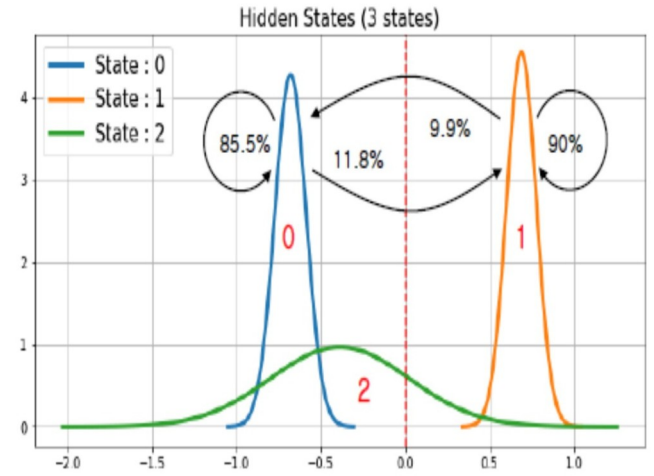
☐ Stock return state-change exercise

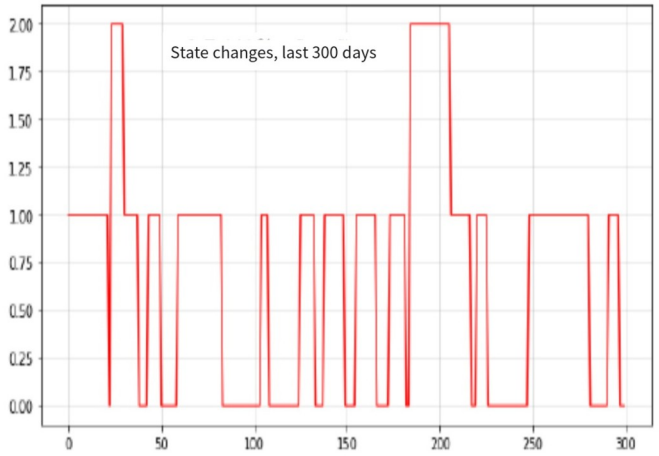
▪ States

State	Mean	Std dev
0	−0.681	0.093
1	0.688	0.088
2	−0.389	0.411

▪ Transition probability

State	0	1	2
0	0.855	0.118	0.027
1	0.099	0.9	0.002
2	0.017	0.051	0.932







05

PART 5 · INFORMED TRADING

An HMM Approach to Information-Based Trading

Kyle model, volume time, PIN and VPIN

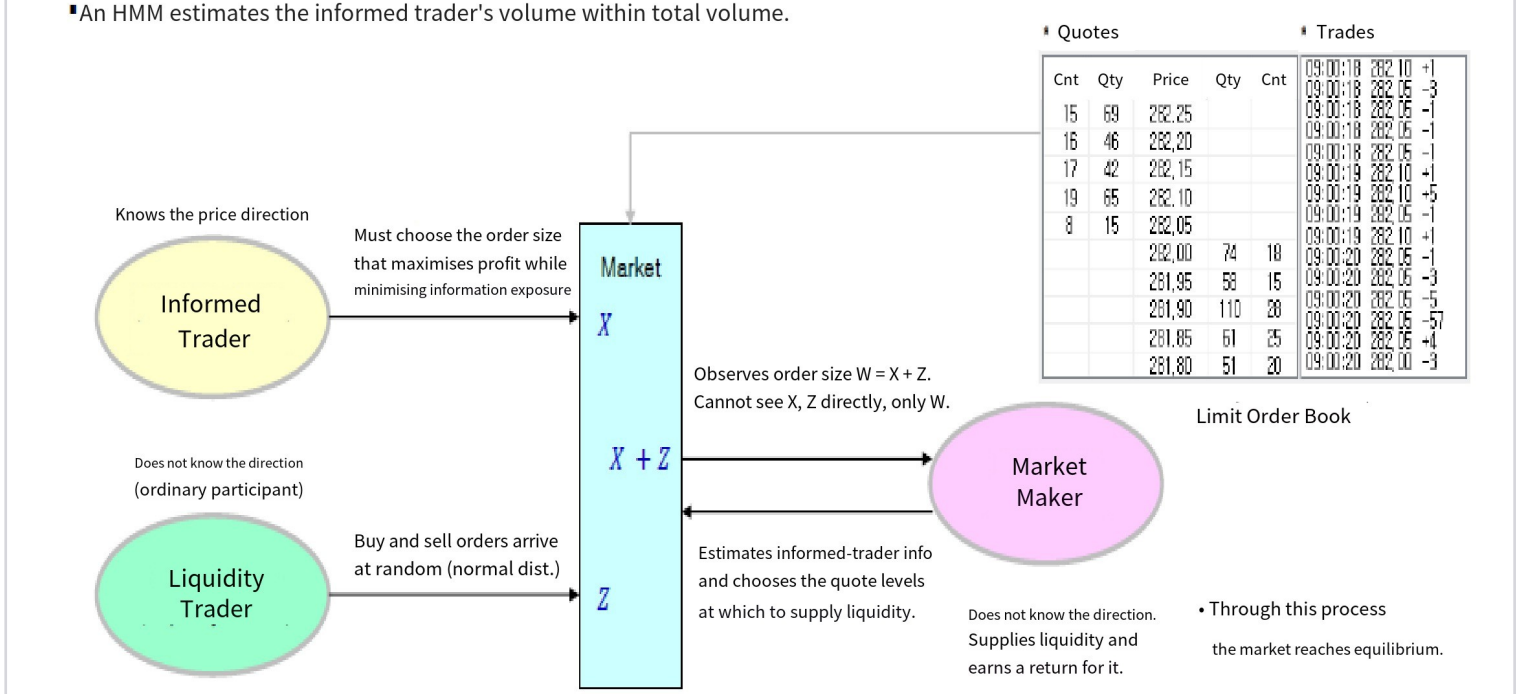
Kyle's model has three players: informed, noise, market maker

Informed trader · liquidity trader · market maker

Players

- Informed: private info
- Liquidity: noise orders
- Market maker: sets price
- HMM: share of informed volume

- A market model with an informed trader, a liquidity trader and a market maker.
- The informed trader holds private information and knows whether the price will rise or fall (an assumption).
- The liquidity trader has no information and trades for its own purposes (hedging) (noise trader). Buy and sell orders arrive at random.
- The market maker posts limit buy and sell orders at the ask and bid at once. These resting orders supply liquidity to the market.
- An HMM estimates the informed trader's volume within total volume.



Tick-time bars adapt to activity where clock-time bars cannot

E-mini S&P 500 futures · clock time vs tick time candles

Clock vs tick

- Clock: fixed time
- Tick: fixed trade count
- Activity spikes after 22:00
- Tick bars stay informative

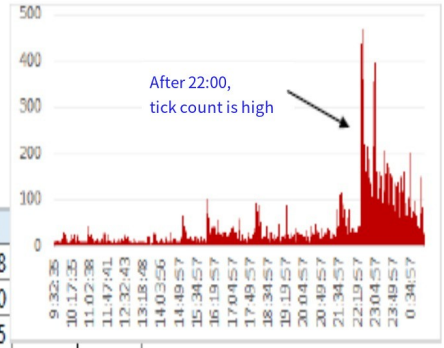
A standard candle chart records prices (open, high, low, close) over a fixed time. Clock time based candle chart.

Market information enters through volume, so all information is reflected in price. The price is a vehicle of information.

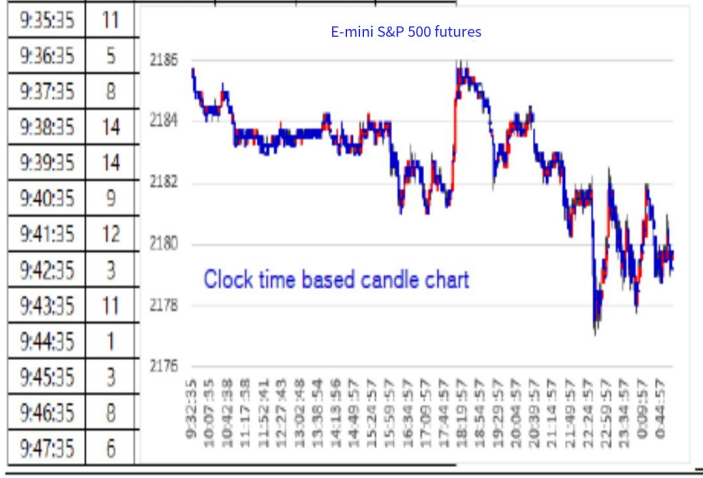
But volume varies by time of day, so each clock-time candle holds a different amount of information.

For example, volume is high just after the open and before the close and low at lunch, so candles formed just after the open and at lunch cannot be treated alike.

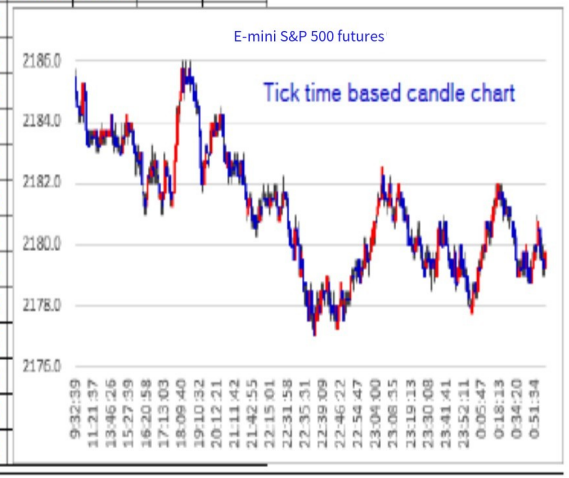
A candle built each time a fixed number of trade ticks occurs is a Tick time based candle chart.



time	ticks	open	high	low	close
9:32:35	8	2185.5	2185.75	2185.5	2185.5
9:33:35	5	2185.5	2185.75	2185.5	2185.75
9:34:35	6	2185.5	2185.5	2185.5	2185.5
9:35:35	11				
9:36:35	5				
9:37:35	8				
9:38:35	14				
9:39:35	14				
9:40:35	9				
9:41:35	12				
9:42:35	3				
9:43:35	11				
9:44:35	1				
9:45:35	3				
9:46:35	8				
9:47:35	6				



time	ticks	open	high
9:32:39	100	2185.5	2185.8
9:43:50	100	2185.0	2185.0
9:55:48	100	2184.5	2184.5
10:01:41	100		
10:14:49	100		
10:23:12	100		
10:31:39	100		
10:42:02	100		
10:54:45	100		
10:57:41	100		
11:03:49	100		
11:12:24	100		
11:21:37	100		
11:32:22	100		
11:41:19	100		
11:50:13	100		



Volume-time bars close after a fixed traded volume

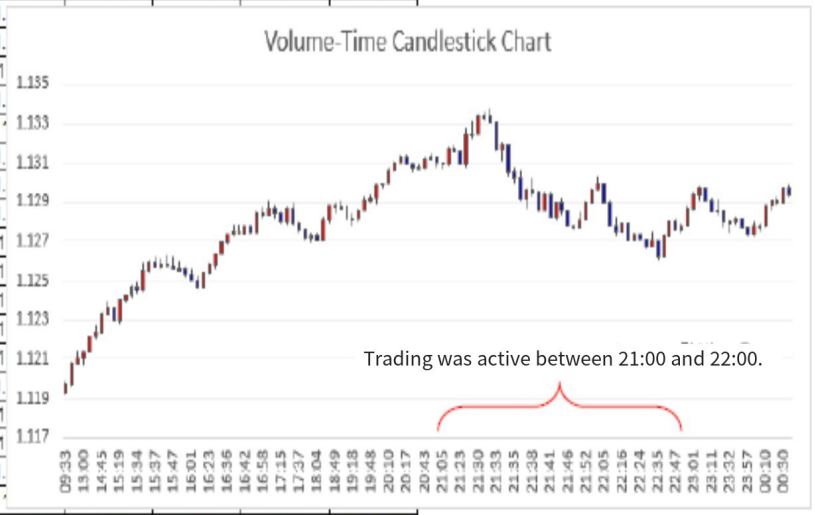
EUR/USD futures, 2016-08-16 · 1,500 lots per bar · buy/sell ratio per bar

Volume time

- Bar = fixed volume
- Record buy and sell
- Ratio B/S per bar
- Input for the HMM

A candle built each time a fixed volume trades is a Volume time based candle chart.

	A	B	C	D	E	F	G	H	I	J	K
1	Euro Currency Futures EUR/USD (2016.8.16 09:32:39 ~ next day 01:08:00)										
2	date	time	ticks	open	high	low	close	volume	buy	sell	buy/sell
3	2016-08-16	09:33	475	1.11925	1.1199	1.1192	1.1197	1500	795	705	1.1276596
4	2016-08-16	11:38	376	1.1197	1.12085	1.1196	1.1208	1500	962	538	1.7881041
5	2016-08-16	12:20	332	1.1208	1.12145	1.12075	1.121	1500	837	663	1.2624434
6	2016-08-16	13:00	280	1.121	1.1214	1.1207	1.12135	1500	956	544	1.7573529
7	2016-08-16	13:58	290	1.12135	1.1222	1.12135	1.				
8	2016-08-16	14:17	396	1.12215	1.12275	1.122	1.				
9	2016-08-16	14:45	388	1.12235	1.12335	1.1223	1				
10	2016-08-16	15:05	238	1.1233	1.12395	1.1233	1.				
11	2016-08-16	15:07	295	1.12365	1.12365	1.123					
12	2016-08-16	15:19	246	1.123	1.12405	1.12285	1.				
13	2016-08-16	15:31	164	1.12405	1.1243	1.1239	1.				
14	2016-08-16	15:33	186	1.12425	1.1248	1.1241	1.				
15	2016-08-16	15:34	219	1.12465	1.1249	1.12435	1				
16	2016-08-16	15:36	159	1.1245	1.1259	1.12445	1				
17	2016-08-16	15:36	280	1.1255	1.12605	1.1254	1				
18	2016-08-16	15:37	290	1.1259	1.12625	1.12565	1				
19	2016-08-16	15:39	327	1.1257	1.12615	1.1256	1				
20	2016-08-16	15:42	288	1.1258	1.12635	1.12565	1.				
21	2016-08-16	15:47	259	1.12575	1.12625	1.1256	1				
22	2016-08-16	15:53	258	1.1256	1.126	1.1255	1				
23	2016-08-16	15:58	231	1.1256	1.1257	1.12515	1.				
24	2016-08-16	16:01	318	1.12535	1.1256	1.125					
25	2016-08-16	16:07	305	1.125	1.12525	1.1246	1.12465	1500	600	900	0.6666667
26	2016-08-16	16:14	344	1.12465	1.1254	1.12465	1.1254	1500	891	609	1.4630542
27	2016-08-16	16:23	215	1.1254	1.1259	1.1253	1.12585	1500	1075	425	2.5294118
28	2016-08-16	16:28	348	1.12585	1.1264	1.1256	1.1264	1500	955	545	1.7522936



An HMM infers the informed-trading share from buy and sell flow

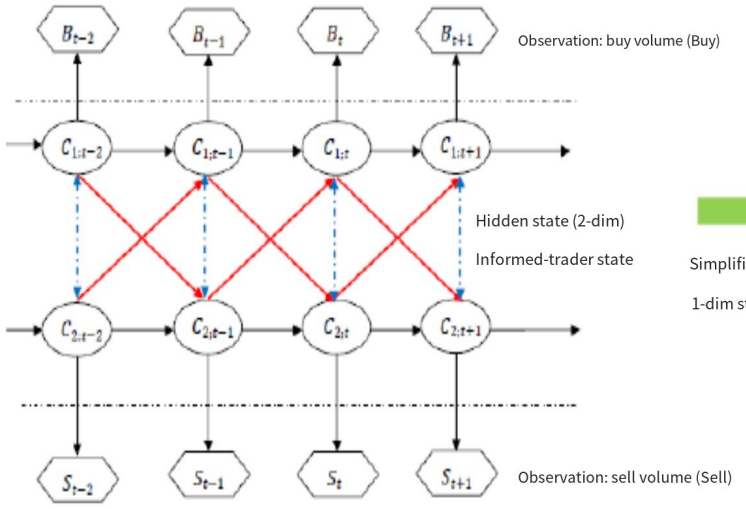
Yin & Zhao (2012), A Hidden Markov Process Approach to Information-Based Trading

Model

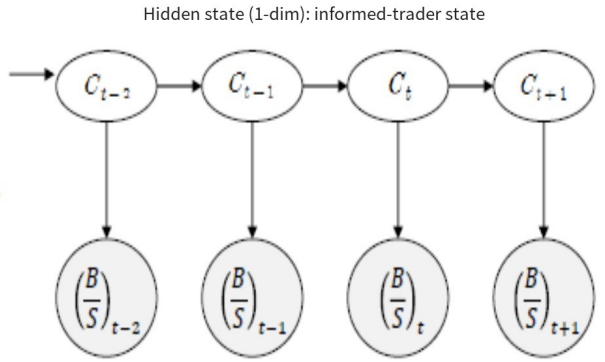
- Hidden: informed share
- Observed: buys, sells
- 2nd order → 1st order
- Observation: B/S

Inferring the informed-trader hidden state (share) with an HMM

- We estimate the informed-trader share by observing buy and sell order volume, or buy order strength (Buy / Sell), flowing into the market.
- The informed-trader share can be seen as a hidden state of order volume. With good news, buy orders should exceed sell orders; with bad news, sell orders should dominate. Without informed traders (no information entering the market), only noise traders remain, so buy and sell orders arrive at random.
- In market microstructure, PIN (Probability of Informed Trading) and VPIN (Volume Synchronized PIN) estimate the informed-trader share, and an HMM can also estimate PIN.



Simplified to 1-dim state



- A high good-news informed share means high buy strength. A low informed share means a high liquidity share, so strength approaches 1 (B ≈ S).

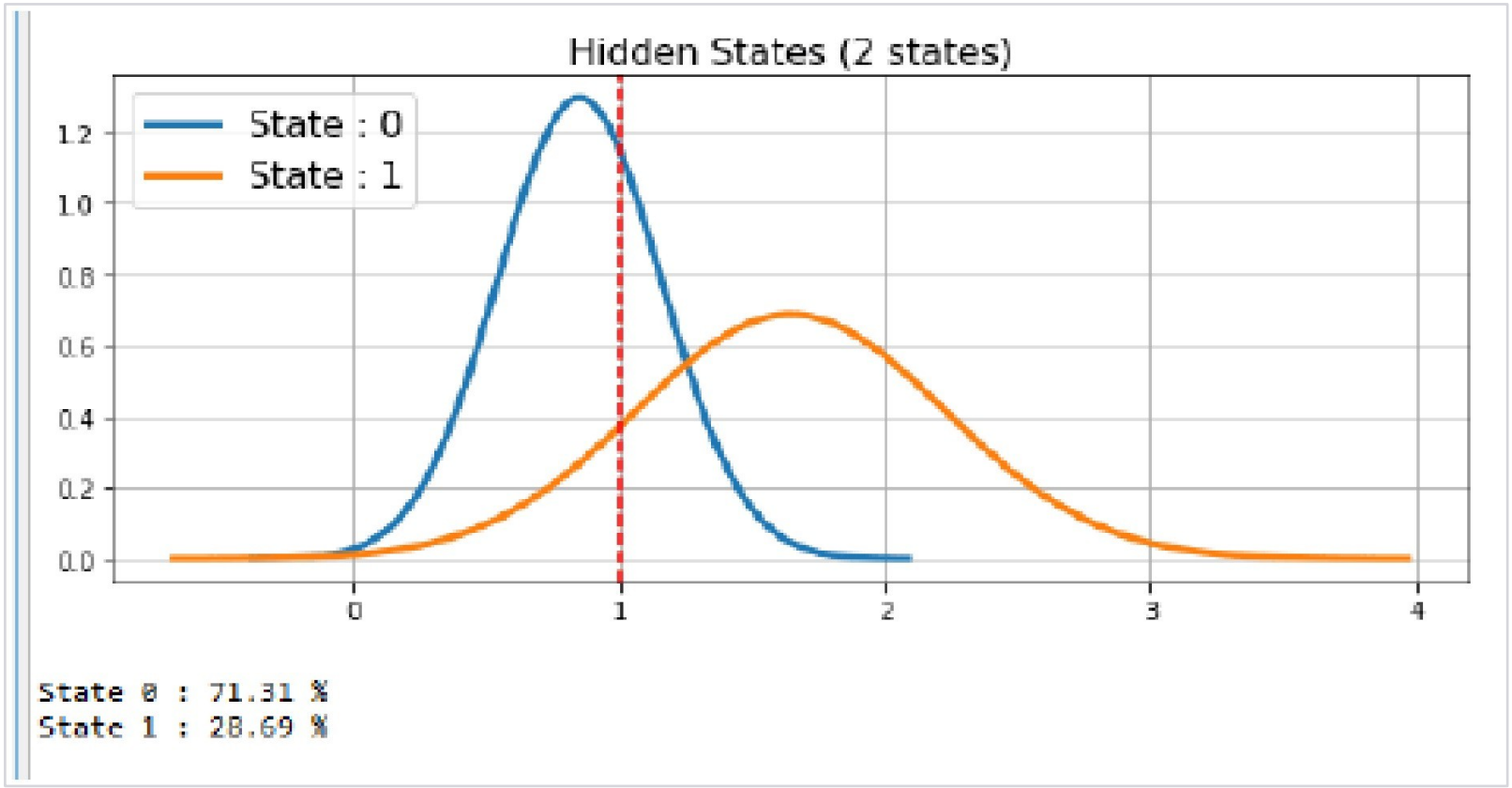
- Source: Xiangkang Yin, Jing Zhao, 2012, A Hidden Markov Process Approach to Information-Based Trading. (Figure 2)

Two states separate ordinary flow from buy-heavy flow

Gaussian HMM on B/S · state 1 (mean B/S > 1) covers 28.69% of bars

Two states

- State 0: 71.31%
- State 1: 28.69%
- Red line: B/S = 1
- State 1 = informed buying



Informed orders appear as short bursts in the state path

Transition: stay 0.841 · switch to informed 0.159 (slide text: 0.84 / 15.9%)

Read the path

- Now: state 0
- Stay prob. 0.841
- Switch prob. 0.159
- Circles: informed bursts

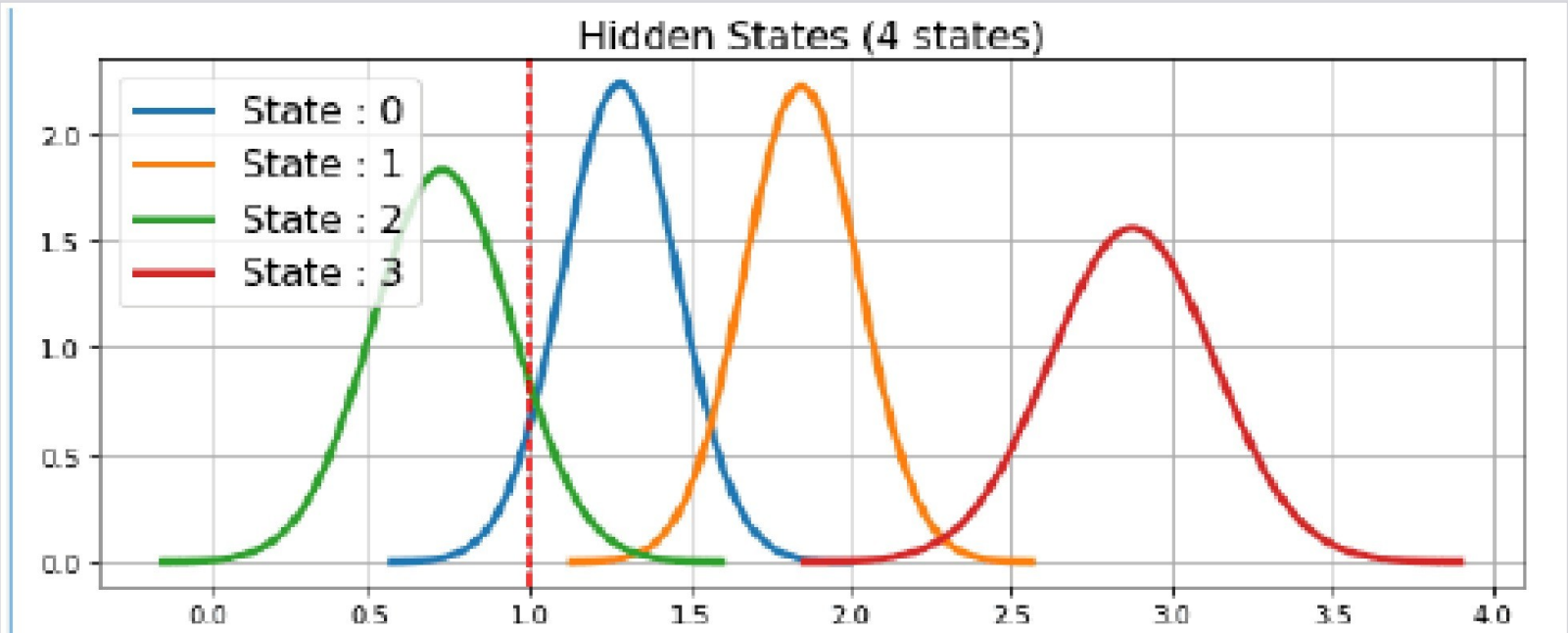


Four states describe order flow without a structural model

No Kyle or PIN assumptions are needed for the HMM

Four states

- State 0: 29.51%
- State 1: 14.75%
- State 2: 51.64%
- State 3: 4.10%

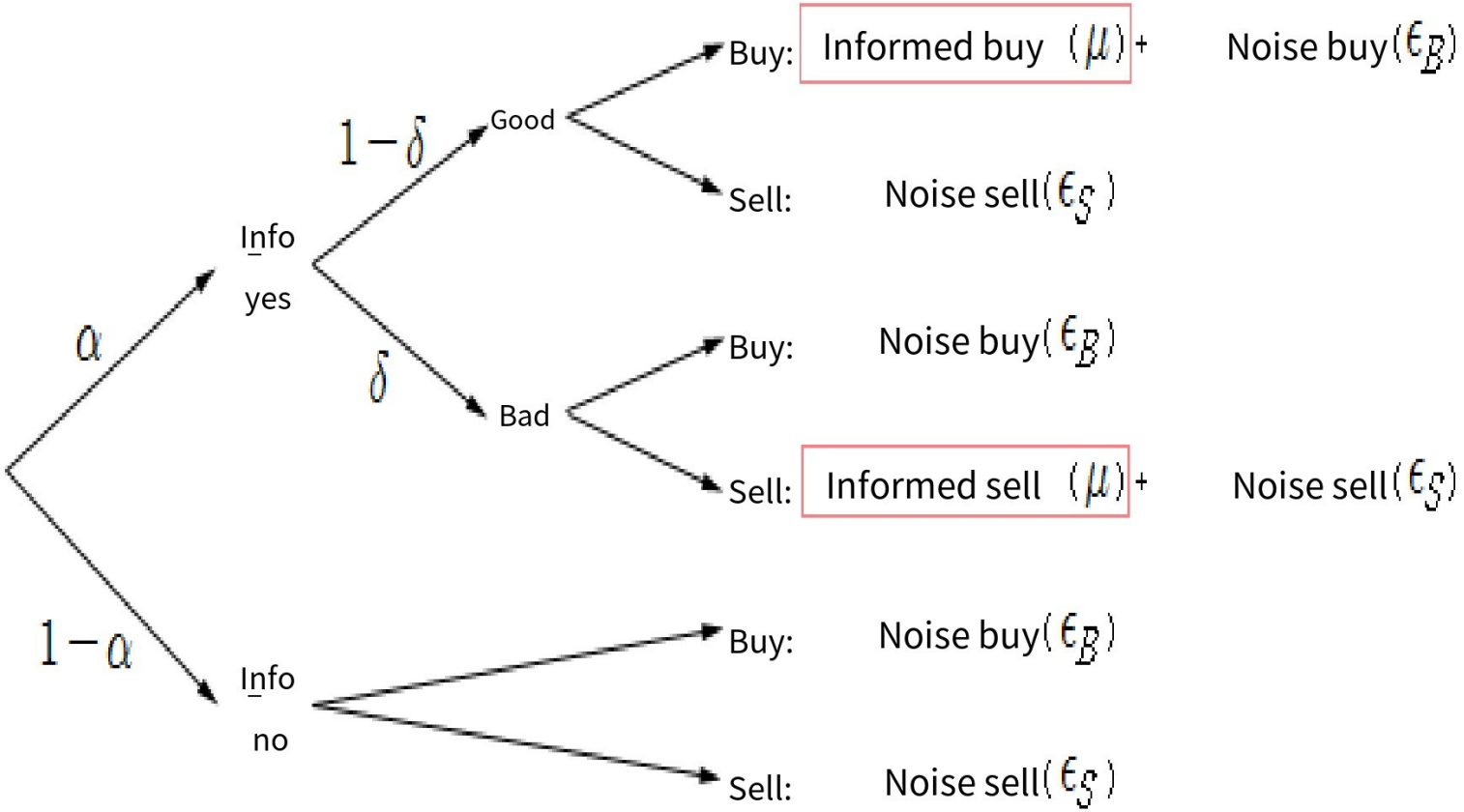


PIN models trade arrivals as a tree of information events

Easley et al. (1996) · information event, good news, bad news

Event tree

- α: info event occurs
- δ: news is bad
- μ: informed arrival
- ε_B, ε_S: noise buys, sells



PIN is the informed share of total trading intensity

Derivation on the original slide: buy, sell and total intensities

Steps

- Buy: α(1−δ)μ + ε_B
- Sell: αδμ + ε_S
- Total: αμ + ε_B + ε_S
- PIN = αμ / total

1) Buy intensity $= \alpha(1-\delta)(\mu+\epsilon_B) + \alpha\delta\epsilon_B + (1-\alpha)\epsilon_B$

$= \alpha(1-\delta)\mu + \epsilon_B$ ← Total buy intensity observed in the market

2) Sell intensity $= \alpha(1-\delta)\epsilon_S + \alpha\delta(\mu+\epsilon_S) + (1-\alpha)\epsilon_S$

$= \alpha\delta\mu + \epsilon_S$ ← Total sell intensity observed in the market

3) Total intensity $= \alpha(1-\delta)\mu + \epsilon_B + \alpha\delta\mu + \epsilon_S$

$= \alpha\mu + \epsilon_B + \epsilon_S$ ← Total buy + total sell intensity

4) Informed total intensity $= \alpha\mu$ ← Informed traders trade only when informed

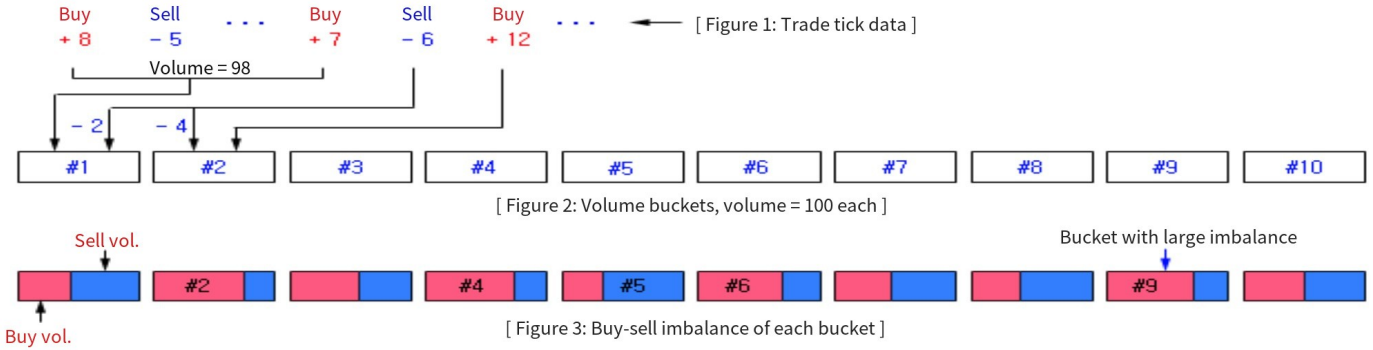
5) $PIN = \dfrac{\text{Informed total intensity}}{\text{Total intensity}} = \dfrac{\alpha\mu}{\alpha\mu + \epsilon_B + \epsilon_S}$

VPIN estimates PIN from buy-sell imbalance in volume buckets

Each bucket holds the same volume V; imbalance |S − B| proxies αμ

Steps

- Group trades in buckets
- Measure |S_i − B_i|
- Average and divide by V
- VPIN ≈ PIN



1) $PIN = \frac{\alpha\mu}{\alpha\mu + 2\epsilon}$

2) $\alpha\mu \approx E[|S - B|] = \frac{1}{n}\sum_{i=1}^{n}|S_i - B_i|$ ← n = Total number of volume buckets; S_i = Sell volume of bucket i; B_i = Buy volume of bucket i

3) $\alpha\mu + 2\epsilon = E[S + B] = V$ ← Bucket volume V is held constant

4) $PIN = \frac{\alpha\mu}{\alpha\mu + 2\epsilon} \approx \frac{1}{nV}\sum_{i=1}^{n}|S_i - B_i| = VPIN$

5) $VPIN = \frac{1}{nV}\sum_{i=1}^{n}|S_i - B_i|$

PIN and VPIN reduce to a ratio of informed to total flow

Formulas from the original slides

$$PIN = \frac{\alpha\mu}{\alpha\mu+\epsilon_B+\epsilon_S} \qquad VPIN = \frac{1}{nV}\sum_{i=1}^{n}|S_i - B_i|$$

PIN

- Needs α, δ, μ, ε by MLE
- Daily buy/sell counts

VPIN

- Volume buckets, no MLE
- Works at high frequency

Order-flow imbalance counts bid and ask changes at the top of the book

Cont et al. (2010) · event-level e_n aggregated over interval k

OFI

- Bid-side event: + q_B
- Ask-side event: – q_A
- Sum e_n over interval
- Another HMM input

Alternative, event-based (Level-1) OFI as per (Cont et al., 2010) is based on order book events:

$$e_n = I_{\{P_n^B \geq P_{n-1}^B\}} q_n^B - I_{\{P_n^B \leq P_{n-1}^B\}} q_{n-1}^B - I_{\{P_n^A \leq P_{n-1}^A\}} q_n^A + I_{\{P_n^A \geq P_{n-1}^A\}} q_{n-1}^A$$

Aggregate over interval k:

$$\mathrm{OFI}_k = \sum_{n=N(t_{k-1})+1}^{N(t_k)} e_n$$

A few statements in the original slides need care

Page numbers refer to the original lecture slides; read the figures with the notes below

Original page	As printed	Note
p.2	E-step finds centres, M-step minimises distance	Usual order: assign points (E), then move centres (M)
p.24	Text: about 7 states	Chart note says 6; both are defensible
p.30	Annual mean −0.681, std 0.093	Same numbers as state-0 μ, σ on p.27; unit not stated
p.38	0.84 and 15.9%	Matrix shows 0.841 and 0.159

DISCUSSION

Four questions link the algorithms to market use

Use the weather example and the KOSPI exercise

Q1	K-Means and an HMM can both split returns into groups. What does the HMM add?	**Clustering**
Q2	Why can the most likely single state each day differ from the Viterbi path?	**Decoding**
Q3	Baum-Welch finds a local optimum. How would you choose the starting θ for KOSPI?	**Learning**
Q4	State 1 of B/S covers 28.69% of bars. Is that a trading signal on its own?	**Order flow**

SUMMARY

One EM idea runs from clustering to market regimes

Group, add time, apply to markets

1

K-Means clusters without order

Assign and update steps already follow the EM pattern.

2

HMM algorithms add the time dimension

Forward sums paths (3.10%), Viterbi keeps the best path (S → R → R → S, 0.2419%).

3

Market data become regimes and signals

KOSPI return states, BIC-based model choice, and informed-flow states from B/S, PIN and VPIN.

Fit on training data, decode on test data, and check the state before acting.